

**A Monte Carlo Pro Forma Valuation Framework for
International Expansion Strategy Across Country Types**

Atharva Mangesh Barad

School of Business, University of Washington Bothell

B BUS 499: Independent Study

Faculty Sponsor: David Shaw

August 21, 2026

Executive Summary

This study develops a 15-year Monte Carlo pro forma valuation framework for evaluating international warehouse-club expansion across the World Bank's four income-group reference classes. Costco Wholesale Corporation is used as the empirical company calibration because its membership-based, high-volume, thin-margin operating model provides a concrete financial system through which country conditions can be translated into revenue, margins, working capital, development spending, financing capacity, and required return. The framework is a research model of a Costco-like six-warehouse expansion; it is not a forecast of Costco's consolidated results or a recommendation to enter any named country.

The final V5 architecture combines 21 Country Screening Tool (CST) factors, two additional foreign-exchange states, a 21x21 Analytic Hierarchy Process (AHP) governance matrix, one 23x23 stochastic dependence matrix, income-group-specific empirical marginal distributions, company residual uncertainty, a temporally evolving country-state system, inflation and currency translation, warehouse-age productivity, construction delay, network support, dynamic annual WACC, terminal value, and strategy-specific financing continuation. Four World Bank income classes are crossed with three rollout policies - Aggressive, Moderate, and Organic - to produce 12 scenarios, each evaluated with 10,000 Monte Carlo paths.

The model preserves a strict distinction between strategic screening and financial transmission. AHP weights represent relative strategic importance, not NPV elasticities. Country variables affect the pro forma only through defined economic channels, while their stochastic co-movement is captured separately through country-matched rank dependence. The resulting framework allows country conditions, company economics, financing constraints, execution timing, and uncertainty to be evaluated inside one auditable valuation system.

The results show a strong development gradient. Low-income reference-class states are unfavorable for the generic six-store experiment: median NPV ranges from -\$268.9 million under Aggressive to -\$42.2 million under Organic, and strict full-project success is at or below 1.5%. Lower-middle-income scenarios improve but remain negative at the median; Aggressive and Moderate strict success reach approximately 17%. Upper-middle-income states form a transition zone: median NPV remains negative, but the favorable tail is substantial and Moderate strict full-project success reaches 38.7%. High-income states are strongest: median NPV is \$257.0 million for Aggressive, \$226.9 million for Moderate, and \$69.6 million for Organic, while strict full-project success reaches 69.4%, 70.6%, and 19.9%, respectively.

Strategy changes the amount, timing, and conditionality of capital commitment. Aggressive guarantees completion of the committed six-store rollout and generally produces the largest mature scale and highest median dollar value in favorable environments, but it requires the most Sponsor / External Capital and creates the widest downside exposure. Moderate uses an initial sponsor-supported stage followed by retained Project Cash and Project Debt capacity; it cuts sponsor exposure materially and produces the highest strict full-project success frequency within every income group, although its advantage over Aggressive is small and payback is slower. Organic protects external capital most strongly, but its retained-cash rule frequently prevents completion of the intended six-store network. In summary: Aggressive pays for certainty with capital. Moderate pays for caution with time. Organic pays for capital preservation with scale.

Table 1
Management Decision Zones From the Final V5 Experiment

Reference class	Decision zone	Model interpretation	Management implication
Low income	Red	Negative median and P90 NPV across all strategies; strict success $\leq 1.5\%$.	Do not advance the generic six-store configuration without exceptional country-specific evidence.
Lower-middle income	Red / Selective	Negative median NPV, but a meaningful positive upper tail.	Advance only exceptional candidates that clearly sit toward the favorable end of the reference class.
Upper-middle income	Yellow / Case-by-case	Negative median NPV with substantial positive upside and ~39% Moderate strict success.	Perform named-country, metro, site, logistics, FX, and financing underwriting before deciding.
High income	Green	Positive median NPV under all strategies and the highest combined success frequencies.	Advance promising candidates to deeper due diligence; do not treat the classification as automatic approval.

Note. Decision zones are conditional interpretations of the calibrated six-warehouse experiment, not real-world probabilities or universal recommendations for every country in an income class.

Research Questions and Principal Findings

1. How can company-specific benefit, cost, and risk indicators be converted into financial assumptions?

By separating strategic importance from financial transmission and mapping each country factor to a defined operating, development, financing, timing, or valuation mechanism.

2. How do World Bank country classifications affect financial risk-return outcomes?

They provide useful reference classes, with simulated economics generally improving from Low- to High-income environments while substantial within-group variation remains.

3. How do investment strategies affect survivability, payback, and return?

Aggressive maximizes scale and completion, Moderate trades slower payback for conditional commitment and lower sponsor exposure, and Organic protects capital but often sacrifices full network completion.

4. Which country-strategy combinations produce the strongest risk-adjusted outcomes?

High-income scenarios are strongest overall: Aggressive produces the highest typical dollar value and scale, while Moderate produces the highest strict full-project success frequency.

Table of Contents

Executive Summary	2
1. Introduction and Research Questions	7
2. Conceptual and Methodological Foundations	10
3. Company, Industry, and Financial Benchmark Analysis.....	13
4. Country Screening, AHP, and Empirical Marginal Distributions	17
5. Monte Carlo Architecture and Calibration	23
6. Rollout and Financing Strategy Design	33
7. Scenario Results and Interpretation	37
8. Robustness, Validation, and Model Refinement.....	48
9. Managerial Recommendations.....	50
10. Limitations, Future Research, and Reflection.....	52
11. Conclusion	56
References.....	57
Appendix A - Model Architecture and Reproducibility	65
Appendix B - CST and AHP Evidence.....	70
Appendix C - Monte Carlo Validation and Diagnostics.....	83
Appendix D - Detailed Scenario Results	89
Appendix E - Data Sources and Provenance	92
Appendix F - AI Use Note	94

List of Tables

Table 1. Management Decision Zones From the Final V5 Experiment	4
Table 2. Costco Calibration and Model Implications	16
Table 3. Final AHP Strategic-Importance Weights	18
Table 4. Country-State to Financial-Transmission Registry.....	25
Table 5. Supporting Return and Payback Metrics Across the 12 Scenarios.....	41
Table 6. Financing Exposure and Execution Outcomes	43

List of Figures

Figure 1. Representative Income-Group Marginal Distributions	21
Figure 2. Final Monte Carlo Architecture With Financing Feedback	23
Figure 3. True-Zero Company-to-Country Bridge	24
Figure 4. Inflation and Foreign-Exchange Transmission Through the Project	29
Figure 5. Final Rollout and Financing Policies	33
Figure 6. Median NPV and P10-P90 Range Across the 12 Configurations	37
Figure 7. Strict Full-Project Success Frequency	39
Figure 8. Warehouse-Level Probability of Opening by Year 15	40
Figure 9. Direct Versus Joint Country-Factor Sensitivity	45

1. Introduction and Research Questions

1.1 Research Problem

International expansion decisions combine strategic attractiveness, operating feasibility, financing capacity, and uncertainty. Country-screening frameworks can help managers identify attractive markets, while financial models can estimate project value after a set of assumptions has been chosen. A gap remains between these two activities: country scores do not automatically explain how external conditions alter a company's revenues, margins, working capital, construction costs, financing availability, timing, and required return.

This study addresses that gap by developing a Monte Carlo pro forma valuation framework that converts a company-specific Country Screening Tool into a stochastic financial model. Rather than terminating in a weighted country score, the analysis asks how each relevant country condition would enter the operating and valuation mechanics of a physical international expansion. The objective is decision support under uncertainty, not a single deterministic ranking.

Costco Wholesale Corporation provides the empirical calibration because its warehouse-club model is financially distinctive and internationally exposed. The company combines recurring membership fees, high warehouse productivity, thin merchandise margins, rapid inventory turnover, meaningful supplier financing, large physical development requirements, and sensitivity to foreign currency. These features create observable pathways through which country conditions can affect project cash flow. Costco is therefore an empirical benchmark for the model, not the subject of a forecast of consolidated corporate performance.

1.2 Research Objective and Scope

The experiment evaluates a hypothetical six-warehouse international expansion over a 15-year explicit horizon. Four World Bank income-group reference classes - Low, Lower-middle, Upper-middle, and High income - are combined with three rollout and financing policies: Aggressive, Moderate, and Organic. The 12 resulting configurations are evaluated with 10,000 Monte Carlo paths each. The same underlying country and economic path is shared across strategies where possible so that strategy comparisons reflect policy differences rather than unrelated random draws.

The framework is intentionally modular. Company operating calibration, country-state evidence, capital markets, business-model transmission, temporal uncertainty, strategy, financing continuation, simulation, and outputs are separate layers. This makes it possible to audit where an assumption enters the model and reduces the risk that country risk is counted repeatedly through overlapping revenue haircuts, cost penalties, and discount-rate surcharges.

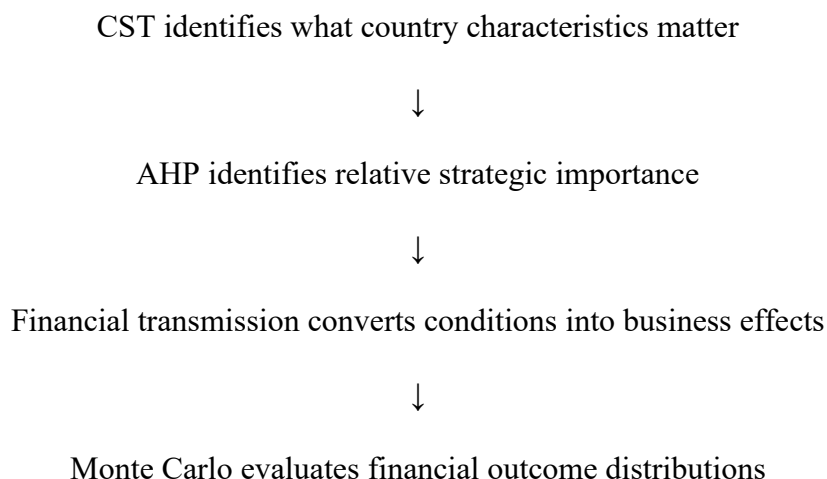
1.3 Research Questions

1. How can company-specific benefit, cost, and risk indicators be converted into financial assumptions for international expansion modeling?
2. How do World Bank country classifications affect financial risk-return outcomes?
3. How do different investment strategies affect survivability, payback timing, and return?
4. Which country-investment combinations produce the strongest risk-adjusted outcomes?

1.4 Analytical Logic

The analysis follows a sequential logic. The Country Screening Tool identifies which external country characteristics matter to the business model. AHP makes their relative strategic importance explicit. The financial-transmission layer then converts those conditions into specific

business effects, and the Monte Carlo engine evaluates the resulting distribution of financial and strategic outcomes.



1.5 Contribution

The principal contribution is methodological. The project demonstrates how strategic country screening and multicriteria decision analysis can be connected to a company-specific pro forma without treating qualitative weights as financial elasticities. It also demonstrates how country variables can be modeled as correlated and temporally evolving states, how financing constraints can feed back into real rollout timing, and how value and strategic completion can be reported separately.

The resulting probabilities are conditional model frequencies under the empirical data, structural priors, financial calibration, and success definitions used here. They should not be interpreted as literal real-world probabilities that Costco will succeed or fail in any country. The model is best understood as an auditable reference-class experiment that can be replaced with named-country and site-specific evidence in future applications.

2. Conceptual and Methodological Foundations

2.1 Country Screening and Multicriteria Decision Analysis

Preliminary foreign-market assessment requires firms to compare markets across economic, institutional, infrastructure, cost, and risk dimensions. Cavusgil et al. (2004) show how country clustering and ranking can support preliminary foreign-market opportunity assessment. Such screening is useful because international investment decisions rarely reduce to one variable such as GDP or tax rate. For a warehouse-club business, market size, household consumption, urban concentration, labor and construction costs, logistics, institutions, inflation, financing risk, and other factors can all affect project feasibility.

The Analytic Hierarchy Process provides a transparent way to govern those competing criteria. AHP converts pairwise judgments into a normalized priority vector while testing the internal consistency of the judgment matrix (Saaty, 2008). Prior research has combined AHP with simulation in foreign direct-investment decision problems, demonstrating that multicriteria screening and uncertainty analysis can be complementary rather than competing approaches (Levary & Wan, 1999).

In this study, AHP is deliberately restricted to strategic governance. The weights answer which CST factors deserve the greatest strategic attention. They do not determine how strongly NPV changes when a factor moves. That separate relationship is specified and validated in the financial-transmission layer.

2.2 Probabilistic Investment Appraisal

Traditional capital budgeting often summarizes uncertainty through a small number of deterministic scenarios. Monte Carlo simulation instead propagates distributions of uncertain inputs through the financial model and produces distributions of outputs such as NPV, IRR,

payback, and terminal value. Prior investment-appraisal research has emphasized the usefulness of Monte Carlo methods when multiple uncertain inputs interact and when a single expected-value forecast obscures downside and upside distributions (Hacura et al., 2001; Savvides, 1994).

The project also distinguishes measurable risk from deeper uncertainty. Knight's (1921) distinction is useful because not every aspect of international expansion can be credibly assigned a finely estimated probability distribution. Where evidence is sufficiently rich, V5 uses empirical country and company data. Where direct identification is unavailable, the model uses visible structural priors and subjects consequential choices to diagnostic or robustness analysis rather than presenting them as estimated facts.

2.3 Staging, Irreversibility, and Financing Continuation

International retail development is partially irreversible: land, construction, preopening expenditures, organizational setup, and market-entry costs are committed before the full quality of the opportunity is known. The value of waiting literature shows why delaying irreversible investment can be economically meaningful when uncertainty is material (McDonald & Siegel, 1986). Staged financing research similarly shows how additional capital can be conditioned on information revealed after earlier investment (Gompers, 1995).

These ideas motivate the distinction among Aggressive, Moderate, and Organic strategies. The model does not calculate a formal real-option value, nor does outside literature calibrate the project-specific financing rules. The literature supports the economic mechanism: commitment and conditional continuation can produce different risk, timing, and capital-exposure profiles even when the underlying operating opportunity is identical.

2.4 Integrated Framework

The conceptual framework therefore combines country screening, AHP governance, company-specific financial transmission, empirical probability distributions, dependence, temporal evolution, financing continuation, and enterprise valuation. This sequence is intended to avoid two common shortcuts: treating a country score as a financial return forecast, and applying an undifferentiated country-risk multiplier after the operating model has already been built. The final architecture instead asks where each country condition belongs in the financial system.

3. Company, Industry, and Financial Benchmark Analysis

3.1 Costco's Warehouse-Club Economics

Costco's operating model is built around high sales productivity, low merchandise markups, paid membership, limited assortment, rapid inventory turnover, and a large-format warehouse footprint. In fiscal 2025, Costco reported net sales of \$269.912 billion and merchandise costs of \$239.886 billion, implying a merchandise gross margin of approximately 11.12%. SG&A was \$24.966 billion, or approximately 9.25% of net sales, while membership fees contributed \$5.323 billion (Costco Wholesale Corporation, 2025).

The membership system is a central stabilizing feature. Costco reported 81.0 million paid members and 145.2 million cardholders in FY2025. Worldwide renewal was 89.8%, compared with 92.3% in the United States and Canada, and Executive members accounted for approximately 73.6% of worldwide sales (Costco Wholesale Corporation, 2025). These economics support treating merchandise revenue, membership productivity, and renewal behavior as distinct but connected parts of the project model.

3.2 Working Capital and Operating Efficiency

The warehouse-club model also relies on unusually efficient working capital. FY2025 inventories were \$18.116 billion and accounts payable were \$19.783 billion (Costco Wholesale Corporation, 2025). This supplier-financed operating structure means that changes in inventory days and payable days can materially affect FCFF even when EBITDA is unchanged. Country conditions can also shape these relationships: institutional quality and macro-financial conditions have been linked to working-capital policy and trade-credit behavior in prior research (Anton & Afloarei Nucu, 2022; Kim & Choi, 2003).

V5 therefore models physical inventory requirements and supplier-payment behavior as separate country-sensitive channels rather than permanently fixing operating working capital at a single percentage of sales. The company-residual dependence robustness comparison is reported in Appendix C, Exhibit C03.

3.3 International Scale and Currency Exposure

Costco operated 914 warehouses at FY2025 year-end, including 285 outside the United States. International operations including Canada represented approximately 27% of net sales and 34% of operating income. The company's average warehouse is approximately 147,000 square feet, making land, construction, equipment, utilities, labor, and logistics directly relevant to physical international expansion (Costco Wholesale Corporation, 2025).

Costco's international footprint also reflects a sustained expansion beyond its U.S. base rather than a purely domestic operating model. Its competitive position rests on the combination of membership economics, limited assortment, high warehouse productivity, purchasing scale, and network density, while competitive intensity and the economics of additional warehouse capacity can differ across markets. The international operating precedent therefore supports using Costco as the focal calibration, while the variation across foreign markets reinforces the need for country-specific financial analysis rather than assuming that the U.S. model transfers unchanged.

Currency exposure is economically meaningful rather than merely accounting presentation. Costco reported that foreign-exchange movements reduced FY2025 net sales by approximately \$1.943 billion and net income by approximately \$97 million (Costco Wholesale Corporation, 2025). This supports modeling inflation, imported-input exposure, expected

currency depreciation, FX volatility, and sponsor-currency translation inside the annual project rather than applying an FX adjustment only after NPV has been calculated.

3.4 Capital Capacity and Sponsor Boundary

FY2025 cash and cash equivalents were \$14.161 billion, short-term investments were \$1.123 billion, operating cash flow was \$13.335 billion, capital expenditures were \$5.498 billion, and long-term debt was approximately \$5.805 billion (Costco Wholesale Corporation, 2025). These figures support the use of a strong-parent valuation proxy and the plausibility of sponsor support in the project experiment.

They do not imply that the sponsor's capital is unlimited. V5 tracks the amount of Sponsor / External Capital required by the project, but it does not calculate Costco's maximum corporate borrowing capacity or optimize the upstream mix of cash, parent debt, or equity used to supply that contribution. Project Debt is defined separately as borrowing supported by the expansion project's own demonstrated EBITDA, cash flow, and debt-service capacity.

3.5 Historical Calibration and Warehouse-Age Productivity

Company-side operating uncertainty is calibrated from FY2016-FY2025 Costco history. Six residual families capture revenue/comparable-sales variation, merchandise-cost ratio, cash SG&A, membership renewal, inventory days, and accounts-payable days. The central sampler preserves recent empirical shape within a bounded P10-P90 interval and recenters the bounded distribution so that it does not introduce unintended drift.

Warehouse maturation is calibrated from Costco FY2025 cohort/calendar evidence using a deterministic 10-year warehouse-age productivity curve. The frozen V5 engine uses a 10-year deterministic True-Zero merchandise curve of \$74.913, \$74.913, \$72.572, \$78.424, \$81.936, \$93.641, \$94.421, \$95.592, \$108.077, and \$111.979 million. The same relative productivity

ratios are applied to mature membership revenue of \$1.734 million. No separate stochastic warehouse-ramp residual is sampled because the available evidence is insufficient to identify one independently. The complete curve and calibration evidence appear in Appendix A, Exhibit A06.

Retail expansion depends on more than the number of stores opened. Customer growth, spatial distribution, network density, and competition affect the amount of economically incremental capacity a market can support (Einav et al., 2021; Holmes, 2008; Lim et al., 2021). V5 therefore separates observed installed density from economic network support and from management's decision to construct another warehouse.

Table 2
Costco Calibration and Model Implications

Observed company evidence	Selected FY2025 calibration	Final model implication
Net sales / merchandise costs	\$269.912B / \$239.886B	Calibrates warehouse-club revenue scale and thin merchandise-margin structure.
SG&A	\$24.966B; ~9.25% of sales	Supports explicit recurring operating-cost architecture.
Membership fees / paid members	\$5.323B / 81.0M	Separate membership-revenue and renewal channels.
Worldwide renewal	89.8%	Company membership-renewal benchmark with residual uncertainty.
Inventory / accounts payable	\$18.116B / \$19.783B	Separate inventory-days and AP-days working-capital channels.
Warehouse network	914 total; 285 outside U.S.	International operating precedent and network-density evidence.
FX impact	-\$1.943B sales; -\$97M net income	Supports explicit annual FX exposure and translation.
CapEx	\$5.498B	Physical-investment benchmark and capital-intensity context.
Cash + short-term investments	\$15.284B	Strong sponsor-capacity context; not an unlimited funding assumption.
Operating cash flow	\$13.335B	Supports strong-parent cash-generation context.
Long-term debt	~\$5.805B	Supports low-leverage parent/proxy capital structure.
Warehouse age/cohort evidence	FY2025 cohort averages	Calibrates audited 10-year warehouse-age productivity curve.

Note. Company values are primarily from Costco Wholesale Corporation (2025). The model uses Costco as an empirical calibration, not as a forecast of consolidated Costco performance.

The complete source and transformation lineage for the company, country, and capital-market inputs is documented in Appendix E, Exhibit E01.

4. Country Screening, AHP, and Empirical Marginal Distributions

4.1 Country Screening Architecture

The country side of the model begins with 21 company-relevant CST factors organized as Benefits, Costs, and Risks. Two additional foreign-exchange variables - Expected FX Depreciation and FX Volatility - are added directly to the stochastic engine because currency behavior is financially important but does not need to receive a separate AHP strategic weight. The final simulation therefore contains 23 stochastic country-state variables.

World Bank income classifications are used as empirical reference classes rather than representative countries. Each group contains heterogeneous observations, so V5 conditions distributions rather than assigning one fixed country score to Low, Lower-middle, Upper-middle, or High income (World Bank, n.d.).

4.2 AHP Strategic Governance

The final AHP matrix contains all 210 unique comparisons among the 21 CST criteria. The judgments are researcher-derived and Costco-informed; they were not elicited from Costco management. The principal eigenvalue is 21.5357, the Consistency Index is 0.02678, and using a Random Index of 1.64 produces a Consistency Ratio of 0.01633, or 1.633%. This is well below the conventional 10% threshold (Saaty, 2008).

The complete 21x21 pairwise comparison matrix is reported in Appendix B, Exhibit BAP01, and the associated consistency diagnostics are reported in Appendix B, Exhibit BAP02.

Table 3
Final AHP Strategic-Importance Weights

Rank	Factor	Category	Weight
1	Financing Risk	Risk	9.861%
2	Market Size	Benefit	9.824%
3	Household Consumption Capacity	Benefit	9.517%
4	Political Stability	Risk	7.147%
5	Institutional Quality	Risk	6.646%
6	Regulatory Quality	Risk	5.994%
7	Urban Market Availability	Benefit	5.892%
8	Labor Cost Pressure	Cost	5.510%
9	Construction Cost Pressure	Cost	4.706%
9	Tax Burden	Cost	4.706%
11	Inflation Risk	Risk	4.197%
12	Expected Market Growth	Benefit	4.138%
13	Construction Delay Risk	Risk	3.853%
14	Site Access Pressure	Cost	2.777%
15	Corruption Risk	Risk	2.543%
16	Metropolitan Customer Concentration	Benefit	2.418%
17	Electricity Cost Pressure	Cost	2.387%
18	Logistics Disruption Risk	Risk	2.356%
19	International Shipping Efficiency	Cost	2.126%
20	Macroeconomic Fragility	Risk	2.045%
21	Labor Underutilization	Cost	1.357%

Note. AHP weights are strategic-importance weights. They are not financial elasticities, causal coefficients, or shares of NPV variance.

At the category level, Risk accounts for 44.64% of total AHP weight, Benefits 31.79%, and Costs 23.57%. This pattern is consistent with a capital-intensive, long-lived foreign-entry decision in which financing, institutional continuity, and downside protection matter strongly alongside demand potential. It remains a researcher-derived, Costco-informed governance profile rather than a universal weighting for international expansion.

The ranking is economically consistent with the warehouse-club problem the AHP was designed to govern. Financing Risk, Market Size, and Household Consumption Capacity form the top tier because the project needs both sufficient demand scale and a financing environment capable of supporting a long-lived physical rollout. Political Stability, Institutional Quality, and Regulatory Quality collectively receive substantial weight because a warehouse network depends on permits, contracts, suppliers, infrastructure, and policy continuity over many years. Labor Cost Pressure, Construction Cost Pressure, and Tax Burden remain material because Costco-like economics operate on thin merchandise margins and require substantial up-front physical

investment. Inflation Risk ranks near the middle rather than the top: persistent inflation can erode cost control and interact with currency depreciation, but its effects are also transmitted through the separate CPI, FX, and financing architecture rather than being counted again as a generic country penalty.

The lower-ranked factors should therefore not be read as unimportant. Some, such as Macroeconomic Fragility and Logistics Disruption Risk, are intentionally narrower or operate strongly through the dependence and Stress systems; giving them additional top-tier AHP weight could double count mechanisms already represented elsewhere. The weight pattern is best interpreted as a governance map of what management should prioritize during screening, not as an empirical ranking of which variables explain the most NPV variation. That distinction becomes visible in Section 7.7, where some moderate-weight factors have large direct financial effects and some high-level institutional factors have much larger joint than isolated effects.

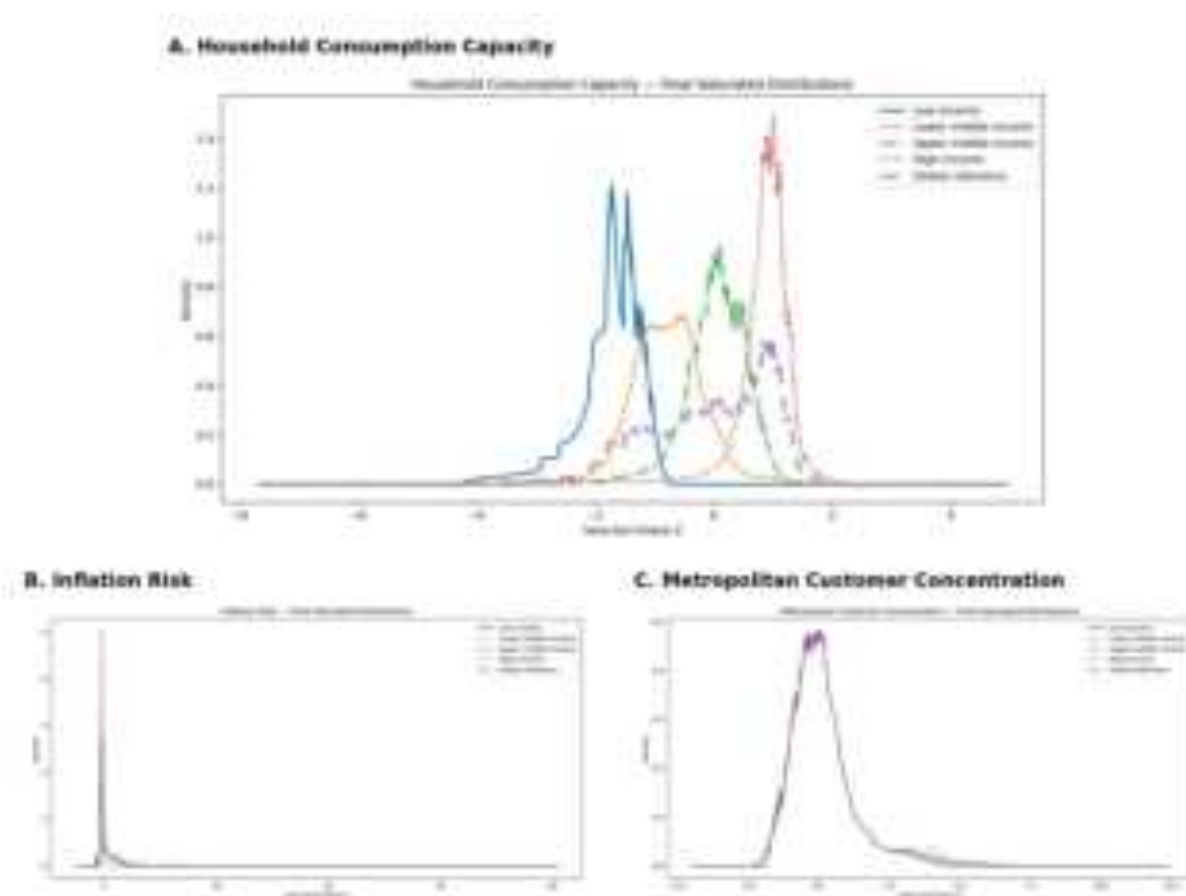
4.3 Empirical Marginal Distributions

Each of the 23 country variables is represented by an income-group-specific empirical marginal distribution rather than a normal distribution imposed for convenience. The final engine uses adaptive/asymmetric kernel estimation where appropriate, cross-validated partial pooling when group samples are limited, a minimum own-group contribution, soft tail saturation, 6,000 numerical grid points, and 100,000 diagnostic draws per factor/group during calibration. Transformations are selected factor by factor before global standardization: eight positive scale variables use the natural logarithm to reduce extreme right-skew, while the remaining fifteen remain on their raw constructed scale. Appendix E, Exhibit E01 reports the exact transformation for every factor.

The distributions reveal three important empirical patterns. Household Consumption Capacity shows clear group separation: its Selected-Z median rises from -1.674 in Low income to -0.770 in Lower-middle, 0.076 in Upper-middle, and 0.924 in High income. Inflation Risk is highly asymmetric, with raw skewness of approximately 7.15 and excess kurtosis of approximately 59.19; the Low-income Selected-Z distribution has a P99 near 9.667. Metropolitan Customer Concentration, by contrast, has almost identical central quantiles across the four income groups, demonstrating that income classification does not partition every business-relevant country characteristic.

The right tail is also economically important for several variables beyond Inflation Risk. In the Low-income calibration, the P99 Selected-Z value is approximately 5.58 for Financing Risk, 8.63 for Expected FX Depreciation, 6.40 for FX Volatility, and 4.80 for Logistics Disruption Risk. Because these variables are coded so that larger values represent more adverse states, the extended right tails represent rare but severe financing, currency, and supply-chain conditions. V5 does not collapse that tail mass into a generic country haircut: Financing Risk changes spreads, Project Debt conditions, accounts-payable behavior, and Stress; Expected FX Depreciation shifts the central currency path; FX Volatility scales annual currency innovations; and Logistics Disruption Risk affects merchandise cost, inventory, and Stress. Preserving these asymmetric tails is therefore a substantive reason for using empirical marginals rather than a common normal assumption.

Figure 1
Representative Income-Group Marginal Distributions



Note. Panels illustrate three distinct empirical behaviors: clear income-group separation (Household Consumption Capacity), strong skew/tail sensitivity (Inflation Risk), and substantial cross-group overlap (Metropolitan Customer Concentration). Complete marginal evidence appears in Appendix B, Exhibit BAP03: Figure B2, Figure B16, and Figure B4 correspond to the three representative panels.

Panel A shows why Household Consumption Capacity is a powerful reference-class discriminator. Household consumption per person is closely related to development level, so the four distributions shift progressively to the right from Low to High income, but they still overlap because income classification does not uniquely determine household spending power. The factor is log-transformed before standardization, which compresses very large raw values and produces a smoother, mound-like Selected-Z shape rather than allowing a few high-consumption countries to dominate the scale. The small local peaks visible within the curves reflect finite

cross-country samples and the empirical density estimator; they should not automatically be interpreted as separate economic regimes. Financially, the separation matters because Household Consumption Capacity directly affects merchandise revenue, membership productivity, and network support, so differences in this distribution can move both per-store economics and the amount of scale a market can sustain.

Panels B and C show why imposing one common normal distribution would lose important information. Inflation Risk is sharply right-skewed because most country-year observations cluster at moderate inflation while a smaller set of episodes reach very high rates; the asymmetric empirical calibration preserves that tail instead of averaging it away. This is economically important because the downside interacts with local costs and currency translation, although the forward-inflation architecture prevents one crisis observation from being compounded mechanically for all 15 years. Metropolitan Customer Concentration behaves differently: the four income groups overlap almost completely because very rich and much poorer countries can each be geographically concentrated around one dominant metropolitan area, while countries at similar income levels can have very different urban systems. Its distribution therefore signals that income group is a useful prior, not a substitute for named-country and metro-level evidence. The soft tail limits used in the numerical grids are simulation guardrails around the observed empirical support, not claims that the real-world variables are physically bounded.

The complete 23-factor marginal-distribution set is reported in Appendix B, Exhibit BAP03. The master source-to-indicator-to-transformation registry is provided in Appendix E, Exhibit E01.

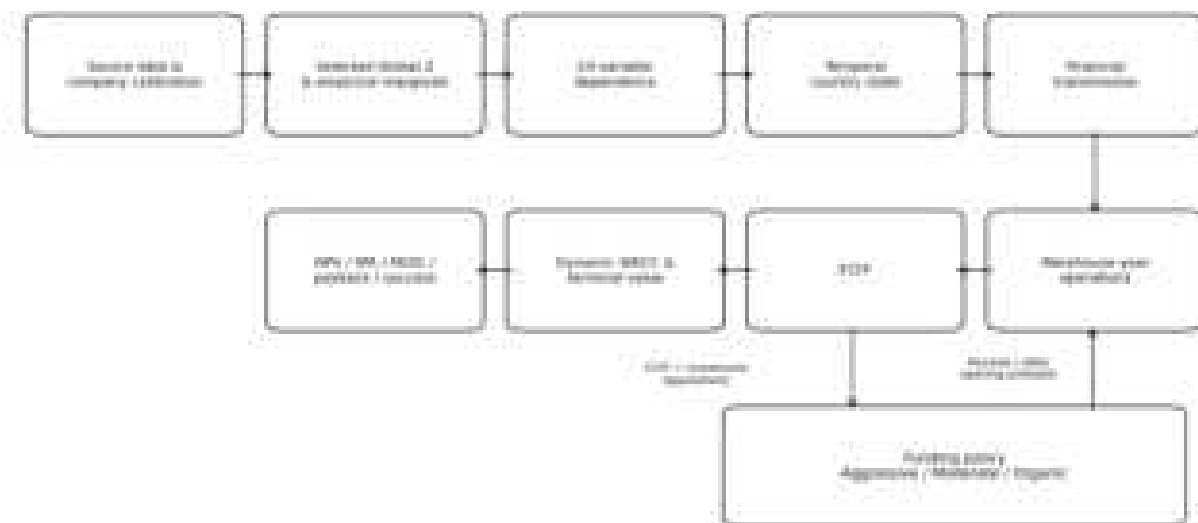
5. Monte Carlo Architecture and Calibration

5.1 Bird's-Eye Architecture

V5 is a variable-first Monte Carlo architecture. Raw company and country evidence are transformed into calibrated drivers; those drivers change specific financial assumptions; financial assumptions evolve year by year; strategy and financing determine the realized warehouse schedule; and the resulting unlevered FCFF stream is valued using annual path-specific capital-market conditions. This sequence is summarized in Figure 2.

Figure 2

Final Monte Carlo Architecture With Financing Feedback



Note. Company calibration and income-group country states enter separate modules before being joined through the financial-transmission layer. Financing can alter the realized warehouse-opening schedule, which then feeds back into operations and future financing capacity.

The architecture is intentionally modular so that observed evidence, derived calibration, and structural priors remain distinguishable. The complete executable equation registry is provided in Appendix A, Exhibit A01, while Appendix A, Exhibit A02 distinguishes observed calibrations, derived values, structural priors, literature-informed priors, and scope boundaries.

5.2 The True-Zero Company-to-Country Bridge

A direct application of destination-country multipliers to observed Costco ratios would risk double counting the U.S. environment already embedded in the company data. V5 therefore uses a 'True Zero' bridge. The model first removes the country-state effect implied by the U.S. calibration from company-side operating relationships, creating a country-neutral benchmark at Selected Global $Z = 0$. The simulated destination country is then applied through the same transmission functions.

Figure 3

True-Zero Company-to-Country Bridge



Note. True Zero is the global statistical center of the selected country-factor transformations. It is not the United States, a risk-free country, a zero-tax state, or an economically ideal market.

The bridge is validated in both directions. With country factors neutralized, the stochastic engine reproduces the country-neutral pro forma. Reapplying the U.S. Selected-Z state reconstructs the original U.S.-calibrated revenue, growth, merchandise cost, SG&A, working-capital, construction, preopening, tax, and spread relationships within pre-specified tolerances. The numerical transition and reconstruction audit is reported in Appendix A, Exhibit A03.

5.3 Country-State to Financial Transmission

The transmission registry is the analytical bridge between strategic country screening and corporate finance. Every stochastic country variable has a defined role. Some variables change revenue productivity, some change physical costs, some affect timing, some influence working

capital, and selected variables affect capital markets or common stress. This prevents the model from turning a weighted CST score into one generic revenue or NPV multiplier.

Table 4

Country-State to Financial-Transmission Registry

Factor	Direct V5 financial role	Interpretation / anti-double-counting boundary
1. Market Size	Network support / economic capacity	Not a direct per-store revenue multiplier.
2. Household Consumption Capacity	Merchandise revenue, membership productivity, network support	Primary direct demand/productivity channel.
3. Expected Market Growth	Real warehouse growth, network-support growth	Changes annual real growth, not initial revenue level.
4. Metropolitan Customer Concentration	Demand accessibility / productivity	Combined with urban availability.
5. Urban Market Availability	Demand accessibility / productivity	Combined with metropolitan concentration.
6. Labor Cost Pressure	Labor SG&A, preopening staffing	Physical recurring and launch labor-cost channel.
7. Labor Underutilization	Preopening staffing/recruitment friction	Separate from wage level.
8. Site Access Pressure	Land/site development cost	Physical development channel.
9. Construction Cost Pressure	Hard-building CapEx	Separate from construction timing.
10. International Shipping Efficiency	MCR, inventory, imported equipment	Merchandise and imported-input efficiency.
11. Electricity Cost Pressure	Nonlabor SG&A	Recurring physical cost channel.
12. Tax Burden	Statutory/effective cash tax	Mapped to tax state plus retained company tax wedge.
13. Political Stability	Common-stress vulnerability and dependence	No separate direct WACC surcharge.
14. Regulatory Quality	Nonlabor SG&A, soft development, preopening administration	Operating/development friction.
15. Macroeconomic Fragility	Common-stress vulnerability and dependence	No standalone direct NPV coefficient.
16. Inflation Risk	Forward CPI process	No direct arbitrary WACC or working-capital-days penalty.
17. Institutional Quality	MCR, nonlabor SG&A, inventory	Operating-friction channel.
18. Corruption Risk / cleanliness state	MCR, SG&A, soft development, preopening	Administrative/operating friction.
19. Logistics Disruption Risk	MCR, inventory, Stress architecture	Distinct from shipping efficiency.
20. Construction Delay Risk	Warehouse-specific 0/1/2-year delay	Executed timing mechanism.
21. Financing Risk	Default spread, CRP, Project Debt conditions, AP, Stress	Principal direct priced country-risk channel.
22. Expected FX Depreciation	Central annual FX path	Separate from volatility.
23. FX Volatility	Scale of realized FX innovation	Separate empirical calibration from Factor 22.

Several anti-double-counting boundaries are deliberate. Market Size affects economic network support rather than directly multiplying each store's revenue. Inflation changes nominal prices and costs through CPI and interacts with FX rather than receiving a second discount-rate penalty. Political Stability and Macroeconomic Fragility help form correlated country states and Stress vulnerability but do not each receive an additional WACC surcharge. Financing Risk is the principal direct priced country-risk channel.

5.4 Company Residual Uncertainty

Country conditions do not explain all operating variability. V5 therefore samples six Costco residual families: revenue/comparable sales, merchandise-cost ratio, cash SG&A, membership renewal, inventory days, and accounts-payable days. The residual history uses FY2016-FY2025, providing ten annual observations per family.

The central sampler uses PERCENTILE.INC-style interpolation with a uniform draw between the historical 10th and 90th percentiles. Each bounded quantile function is recentered by its exact bounded empirical mean so that the residual does not create unintended drift. Year 0 residuals are zero. The P10-P90 interval is a modeling bound, not a confidence interval or a claim that future Costco outcomes cannot exceed the historical range.

Because ten observations are insufficient for a defensible six-dimensional company copula, the central model samples the residual families independently. A robustness mode instead samples winsorized historical-year blocks, preserving observed within-year co-movement without estimating an unstable covariance system. The robustness comparison between independent residual sampling and historical-year block sampling is reported in Appendix C, Exhibit C03.

5.5 Stochastic Dependence and Correlation

The 23 country variables are not simulated independently. For each factor pair, V5 matches countries by ISO3 code and estimates a Spearman rank correlation using the shared observations. Shared sample size, source-period coverage, and year-gap diagnostics are preserved because the underlying international databases are not perfectly synchronized.

Each empirical Spearman coefficient is converted to a latent Gaussian correlation using the standard Gaussian-copula relationship. The assembled pairwise matrix was not positive

semidefinite, so the nearest-correlation procedure of Higham (2002) is used to produce a valid 23x23 matrix. Higham supplies the numerical repair method; the economic correlations themselves come from the country data.

Bootstrap resampling is used as a stability diagnostic rather than as a replacement for the empirical coefficient. The final matrix is then validated end-to-end with 50,000 joint draws per income group after inverse-CDF transformation. One core dependence matrix is used across all four groups, while each group retains its own empirical marginals. This is a parsimonious approximation: dependence may differ structurally by development level, and a Gaussian copula does not capture every form of crisis-specific tail dependence.

The final 23x23 stochastic dependence matrix is reported in Appendix B, Exhibit BAP04, with shared-sample, bootstrap, PSD-repair, and end-to-end validation evidence in Appendix B, Exhibit BAP05.

5.6 Temporal Country-State Architecture

A 15-year expansion requires more than a Year-0 country draw. At the same time, the available evidence does not justify estimating 23 independent bespoke time-series models. V5 therefore groups country variables into temporal classes that distinguish structural scale, persistent real costs, deep institutional conditions, cyclical states, tax reform, risk/stress states, inflation, construction-delay risk, expected FX depreciation, and FX volatility.

Structural-scale and persistent real-cost factors retain substantial country identity over the horizon, while cyclical, risk, inflation, construction-delay, and FX states are allowed to evolve more strongly. Tax can reform discretely and the risk system can enter a temporary Stress regime when Political Stability and Macroeconomic Fragility are jointly adverse. Exact persistence

coefficients, reform probabilities, and Stress multipliers are reported once in Appendix A, Exhibit A02 rather than repeated in the Main Text.

Stress changes the behavior of existing risk, inflation, and FX innovations rather than imposing a second revenue haircut, WACC add-on, CapEx multiplier, or forced devaluation. This treatment is consistent with evidence that sovereign and financial risks can transmit jointly, supporting a correlated stress mechanism rather than multiple independent country-risk penalties (Gilchrist et al., 2021). This keeps the financial effect inside the same transmission channels used in normal states. The synchronized Stress ablation is reported in Appendix C, Exhibit C03.

The anchored latent process preserves part of the initial country identity instead of forcing every dynamic variable back to the global center. Annual innovations continue to use the common dependence structure, while heterogeneous persistence allows relationships to attenuate over time. The exact temporal equations are retained in Appendix A, Exhibit A01.

5.7 Inflation, Foreign Exchange, and Sponsor-Currency Translation

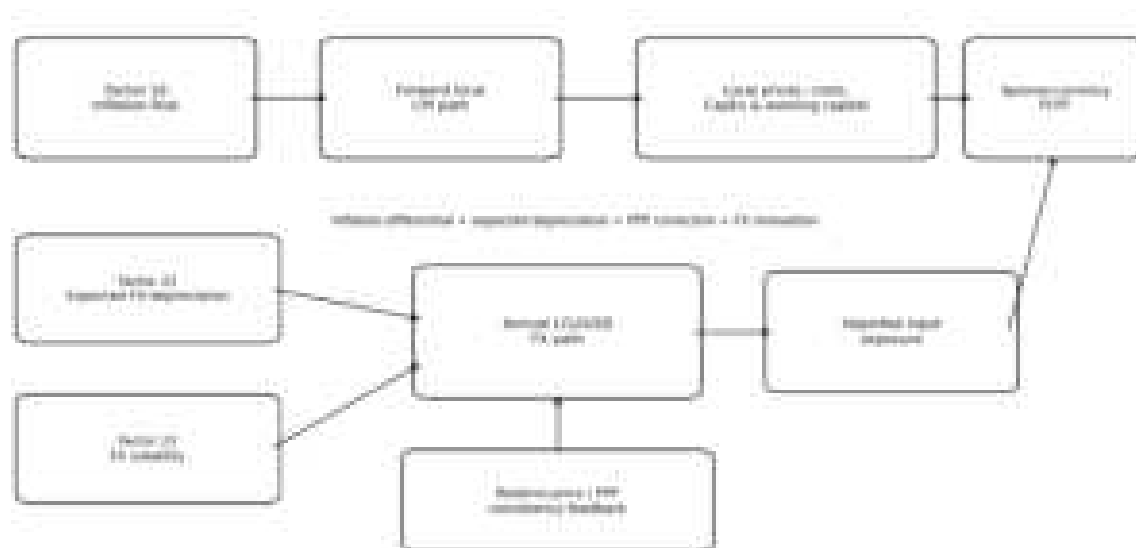
Inflation, expected currency depreciation, and FX volatility are modeled as separate states. Factor 16 supplies the current inflation environment and forward inflation uncertainty. Factor 22 is calibrated from historical annual simple LCU/USD depreciation, while Factor 23 is calibrated separately from annualized monthly FX log-return volatility. Treating FX volatility as a persistent state rather than independent year-to-year noise is consistent with evidence that exchange-rate volatility exhibits meaningful persistence over time (Andersen & Bollerslev, 1996). They are therefore not the mean and standard deviation of one jointly estimated lognormal FX process.

The final forward inflation architecture separates current cross-sectional inflation evidence from the 15-year central path. This separation is also consistent with evidence that

inflation persistence is uncertain and that inflation dynamics can differ materially between low- and high-inflation regimes (Borio et al., 2023; Brandão-Marques et al., 2024). The central forward anchors are 4.0% for Low income, 3.5% for Lower-middle, 3.0% for Upper-middle, and 2.0% for High income. These are literature-informed model anchors, not direct IMF forecasts for the World Bank groups. The IMF World Economic Outlook provides forward macroeconomic context, while the 2% High-income anchor is consistent with the Federal Reserve's longer-run objective (Federal Open Market Committee, 2026; International Monetary Fund, 2026).

Figure 4

Inflation and Foreign-Exchange Transmission Through the Project



Note. Inflation, expected depreciation, and FX volatility remain separately calibrated. FX affects imported operating and development costs as well as sponsor-currency translation.

The exchange-rate system also contains a gradual purchasing-power-parity consistency correction. Accumulated relative price divergence feeds back into subsequent nominal FX movement at 15% per year. Frankel and Rose (1995) support the mechanism of gradual PPP mean reversion, but the 15% coefficient is a literature-informed structural prior rather than a

Costco-specific estimate. The model retains empirical FX volatility and does not force the exchange-rate index back to one.

Imported merchandise, retail pricing, and imported capital expenditures receive distinct pass-through assumptions so that neither complete pass-through nor zero pass-through is imposed universally. A project can therefore perform well in local operating terms and still disappoint in sponsor currency. Appendix C, Exhibit C04 reports the inflation, CPI, FX, and real-exchange-rate diagnostics, while Figure C5 provides a tail-state diagnostic showing how unusually high CPI and FX conditions can produce strong local nominal revenue without necessarily translating into equally strong sponsor-currency value.

5.8 Capital Markets, Dynamic WACC, and Terminal Value

The valuation layer combines the Federal Reserve Gürkaynak-Sack-Wright zero-coupon Treasury term structure, a 4.23% mature-market equity risk premium, a 0.91 Costco-like beta proxy, destination country risk, and a strong-parent debt spread/capital-structure proxy (Board of Governors of the Federal Reserve System, n.d.; Damodaran, 2026b, 2026c; Gürkaynak et al., 2006). The retained corporate debt-spread anchor is approximately 0.79% (ICE Data Indices, LLC, n.d.).

Factor 21 is the principal direct priced country-risk channel. The final country-risk premium is approximately 1.523 times the destination default spread in the central Damodaran calibration (Damodaran, 2026b). Political stability, institutional quality, regulatory quality, and macro fragility are not each stacked as separate WACC penalties because they already affect operations, dependence, or Stress and are related statistically to the financing-risk state.

For reproducibility, the exact frozen cost-of-equity equation, including the 0.23% U.S. reference constant used by V5, is reported in Appendix A, Exhibit A01. The 0.23% term is

treated strictly as the executed V5 reference constant and is not interpreted as a separately estimated U.S. country-risk premium.

Annual WACC values perform the valuation. Each year's FCFF is discounted through the cumulative sequence of path-specific WACCs. V5 separately reports an Effective WACC equal to the geometric annual equivalent of the 15-year WACC path so that IRR can be compared with a single interpretable hurdle. Effective WACC is therefore a reporting diagnostic, not the literal single rate used to discount every annual cash flow.

Terminal value represents continuation of the mature operating network beyond Year 15, not a simulation of new stores after the explicit horizon. The terminal calculation normalizes mature operating relationships, aligns maintenance depreciation with maintenance CapEx, uses a late-horizon stabilized WACC, and applies terminal growth of 2.5% subject to a minimum one-percentage-point WACC-minus-growth guardrail. The implied EV/EBITDA ratio is a DCF reasonableness cross-check only; it is not used to set terminal value.

5.9 Construction Delay and Network Support

Construction delay is modeled at the individual-warehouse level and maps Factor 20 directly into discrete 0-, 1-, or 2-year outcomes. At the neutral state, the executed probabilities are approximately 80% for no delay, 16% for a one-year delay, and 4% for a two-year delay. The exact logistic mapping is reported in Appendix A, Exhibit A01, while the executed construction-delay diagnostic is reported in Appendix C, Exhibit C03.

World Bank FMIS evidence is retained in Appendix E as development/non-executed provenance, but the frozen V5 construction-delay mechanism is calibrated directly from Factor 20 and does not use FMIS to determine positive delay duration.

Delay shifts the actual opening clock and therefore development spending, preopening expenditures, depreciation, revenue, network count, FCFF, and terminal economics. It is not represented as an arbitrary negative revenue haircut. For value-destroying warehouses, delay can even improve NPV by postponing negative capital deployment, which is why the model treats delay through real cash-flow timing.

The network layer starts from a legacy U.S. density reference of 3.5732 warehouse-equivalent units and applies a 1.50x 15-year headroom prior, producing central economic support of 5.3598 before path-specific Market Size and Household Consumption adjustments. The 1.50 multiplier is a structural horizon prior, not a coefficient estimated by Costco, Holmes (2008), Lim et al. (2021), or Guler (2018).

Warehouse rank is compared with economic network support. V5 imposes no productivity penalty while the network remains at or below support; once planned scale exceeds support, marginal productivity declines smoothly, reaching 75% at twice the supported scale. The exact post-support function is reported in Appendix A, Exhibit A01, while the network-headroom sensitivity is reported in Appendix C, Exhibit C03.

Sensitivity around 1.30x, 1.50x, 1.60x, and 1.80x headroom did not overturn the broad decision zones. Network support therefore matters economically without becoming decision-dominant for the defined six-store experiment.

6. Rollout and Financing Strategy Design

The country and operating engine determines what the project could earn; the financing layer determines whether and when management actually builds the intended network. The three strategies are evaluated on comparable country/economic paths but impose different commitments and continuation rules. Financing changes enterprise value through real execution timing rather than by replacing unlevered FCFE with levered cash flow.

Figure 5

Final Rollout and Financing Policies



Note. Sponsor / External Capital is the report term for the project-level funding bucket historically labeled Parent Capital or Parent Funding in portions of the executable. Upstream sponsor financing mix is outside V5.

6.1 Aggressive: Committed Sponsor-Supported Expansion

Aggressive plans all six warehouses for Year 1. Project Cash is used first and Sponsor / External Capital fills any remaining development or liquidity gap. Financing therefore does not cancel or defer the committed rollout; only the warehouse-specific construction process can move a realized opening date. The strategy isolates the economics of rapid commitment: strong states reach scale quickly, while weak states expose more capital before management receives the same opportunity to learn from early operating performance.

The Aggressive policy corresponds to the proposal's front-loaded J-curve concept: heavy early capital deployment precedes the cash generation of a mature network, so cumulative cash flow can be deeply negative before later operating scale has time to recover the investment (Hamilton Lane, n.d.). The J-curve is used here as strategy framing, not as an external calibration of Costco-specific timing, capital requirements, or returns.

The absence of Project Debt in the Aggressive project-level funding table does not mean that Costco corporate uses no debt. V5 does not model whether the sponsor funds its contribution with cash, corporate borrowing, equity, or another source. The category describes funding from the expansion project's perspective.

6.2 Moderate: Staged Expansion With Earned Project Debt Capacity

Moderate plans three warehouses for Year 1 and three additional warehouses for Year 5. The first stage is sponsor-seeded as needed. Later warehouses must be supported by retained Project Cash and rolling Project Debt capacity based on demonstrated operating performance. This structure implements the economic logic of staged irreversible investment without claiming to calculate a formal real-option value (Gompers, 1995; McDonald & Siegel, 1986).

The central borrowing-capacity policy requires $\text{Debt} / \text{normalized EBITDA} \leq 3.0x$, $\text{normalized EBITDA} / \text{cash interest} \geq 6.0x$, and $\text{normalized maintenance cash flow} / \text{Debt} \geq 15\%$. Total capacity is the most restrictive of the three tests. These are researcher-defined analytical boundaries anchored to recognizable corporate-credit concepts; they are not universal bank covenants or thresholds attributed to S&P Global Ratings (2024).

Project Debt behaves as an annually repriced rolling facility. Debt is drawn only when required for growth spending, positive retained cash can repay outstanding debt, and borrowing headroom is reassessed each year. If reassessed capacity falls below debt already outstanding,

V5 does not erase the liability; new headroom simply becomes zero until capacity recovers. If retained cash plus incremental debt headroom cannot finance the next warehouse, that warehouse is deferred one year and reassessed under the next annual operating state. The proceed-or-defer mechanism is also consistent with evidence that credit availability and creditor control rights can constrain corporate investment before outright default or bankruptcy occurs (Nini et al., 2009; Sufi, 2009).

6.3 Organic: Retained-Cash-Funded Expansion

Organic plans one warehouse per year from Years 1 through 6. The initial warehouse receives Sponsor / External Capital as needed, but later warehouses rely on retained Project Cash. There is no Project Debt and no continuing sponsor contribution for later growth. An unfunded warehouse is deferred, and subsequent warehouses cannot leapfrog it.

Organic therefore protects external capital by preserving the option not to expand when the operating project cannot fund the next stage. The same discipline can prevent the project from achieving the intended six-store network, so capital-efficiency metrics must be interpreted together with completion.

6.4 Financing and Enterprise Valuation Boundary

Enterprise valuation remains unlevered. FCFF is discounted using the annual path-specific WACC architecture described in Section 5.8. Sponsor contributions, Project Debt draws, interest, and repayments are tracked separately for financing continuation. They affect enterprise value only when they change the physical opening schedule and therefore the timing and scale of future operations. This avoids deducting financing costs from FCFF while also pricing financing through WACC.

Deterministic financing-aware pro forma evidence for all three policies is reported in Appendix A, Exhibit A04 and is used to verify the executed continuation mechanics before stochastic interpretation.

6.5 Outcome Definitions

Strategic completion means all six planned warehouses are operating by the end of Year 15. Survivability refers to project funding continuity and completion of the intended rollout; it is not the probability that Costco becomes bankrupt. Financial success begins with positive NPV. Strict financial success additionally requires IRR to exceed the path-equivalent Effective WACC.

$$\text{Strict full-project success} = (6/6 \text{ warehouses by Y15}) \text{ AND } (NPV > 0) \text{ AND } (IRR > \text{Effective WACC})$$

This combined measure prevents a fully executed but value-destructive project from being classified as successful and prevents a profitable but incomplete Organic path from being treated as equivalent to a completed six-store expansion.

For interpretation, the simulation therefore produces four economically distinct outcome states: a completed six-store rollout with positive NPV; a completed rollout with negative NPV; an incomplete rollout that nevertheless creates positive enterprise value; and an incomplete rollout with negative NPV. The third state is especially important for Organic: partial execution can be economically positive without satisfying the six-store strategic objective.

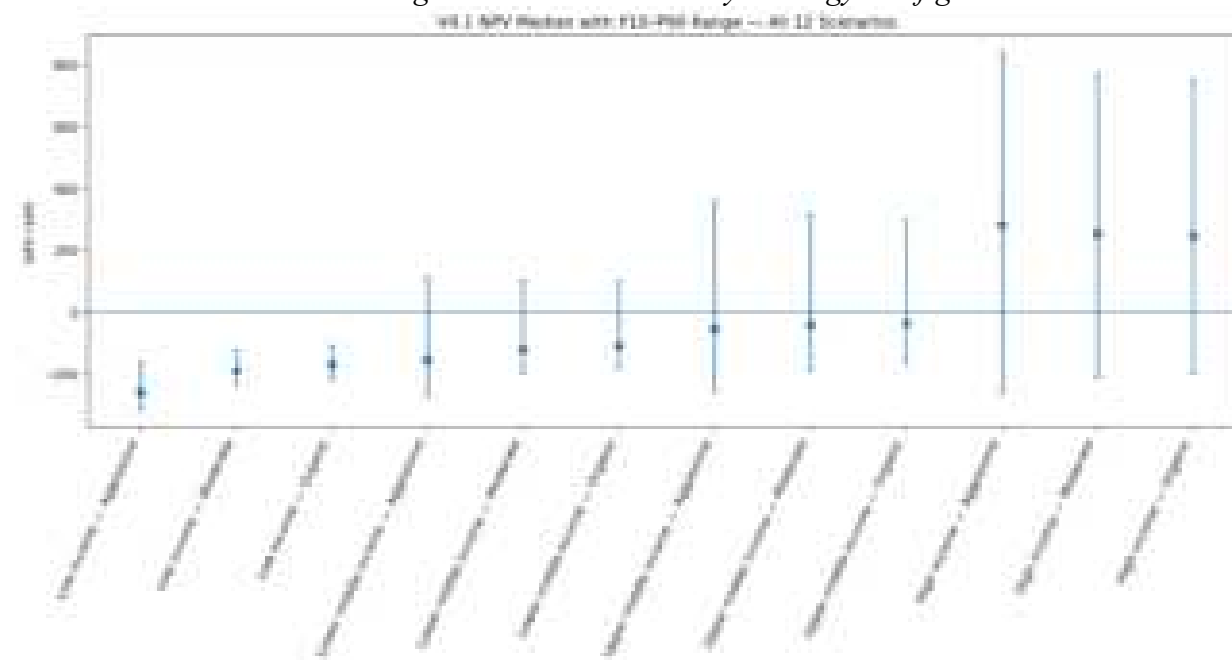
7. Scenario Results and Interpretation

The final simulation evaluates 12 configurations formed by four World Bank income-group reference classes and three rollout/financing strategies. Each configuration contains 10,000 paths. The results should not be interpreted through one metric alone: NPV measures dollar value creation, IRR and MOIC measure return characteristics, payback measures capital duration, financing exposure measures sponsor burden, and completion measures whether the intended physical strategy was actually executed.

7.1 Value Creation, Downside, and Upside

Figure 6

Median NPV and P10-P90 Range Across the 12 Country-Strategy Configurations



Note. Monetary values are millions of U.S. dollars. P10-P90 intervals are simulation percentiles, not confidence intervals.

The dominant result is a clear progression from strongly negative Low-income economics toward positive High-income economics. Low-income median NPV is -\$268.9 million under Aggressive, -\$139.8 million under Moderate, and -\$42.2 million under Organic. Even P90 NPV remains negative for all three policies, while positive NPV occurs in only about

1.4%-1.7% of paths. The generic six-store configuration therefore belongs in a Red reference-class decision zone.

Lower-middle-income economics improve substantially but remain negative at the median: -\$166.7 million, -\$107.2 million, and -\$28.1 million for Aggressive, Moderate, and Organic, respectively. Unlike Low income, the upper tail crosses into positive value, with P90 NPV of approximately \$107.2 million under Aggressive, \$90.9 million under Moderate, and \$23.0 million under Organic. These are exception-driven rather than typical opportunities.

Upper-middle-income states form the clearest transition zone. Median NPV remains negative at -\$71.3 million under Aggressive, -\$50.0 million under Moderate, and -\$14.2 million under Organic, but the P90 rises to approximately \$353.7 million, \$302.9 million, and \$120.7 million. Positive NPV frequencies approach 38%-39%. The wide distribution supports a Yellow / case-by-case interpretation because named-country and site evidence can plausibly move the decision materially.

High-income states produce positive median NPV under every strategy: approximately \$257.0 million for Aggressive, \$226.9 million for Moderate, and \$69.6 million for Organic. Positive NPV occurs in 69.8%, 70.7%, and 74.2% of paths, respectively. Downside nevertheless remains meaningful: P10 NPV is negative for all three. High income is therefore a Green screening signal for deeper due diligence, not automatic approval.

Strategy changes dispersion as well as the center. Aggressive generally produces the largest median NPV and greatest upside in favorable environments, but it also exposes substantially more capital to weak states. Organic compresses both upside and downside by withholding later investment when retained project cash is insufficient. Moderate usually falls

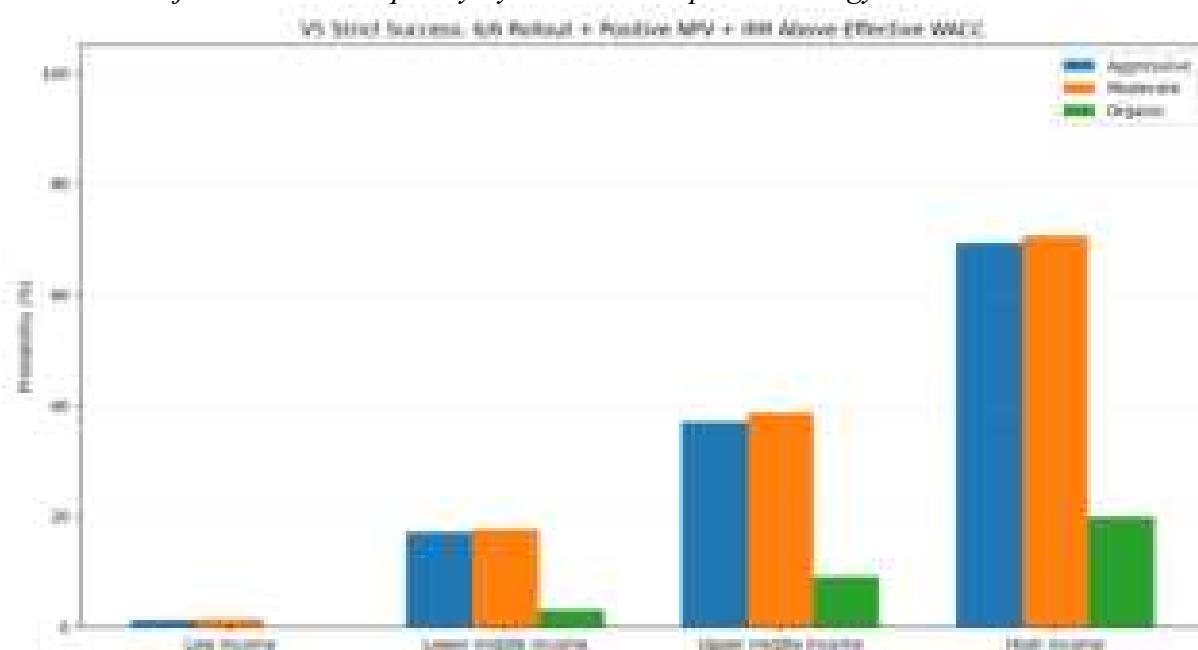
between these extremes. Complete NPV and return distributions are reported in Appendix D, Exhibit D01.

Mean and median NPV do not always rank policies identically. In High income, Moderate's mean NPV is slightly higher than Aggressive's (approximately \$261.8 million versus \$250.6 million) even though Aggressive has the higher median and larger upper tail. The pattern is consistent with staged continuation filtering some severe downside outcomes while sacrificing some typical scale and upside exposure.

7.2 Strict Full-Project Success

Figure 7

Strict Full-Project Success Frequency by Income Group and Strategy



Note. Strict full-project success requires all six warehouses operating by Year 15, NPV > 0, and IRR > path-equivalent Effective WACC. Frequencies are conditional on the V5 calibration.

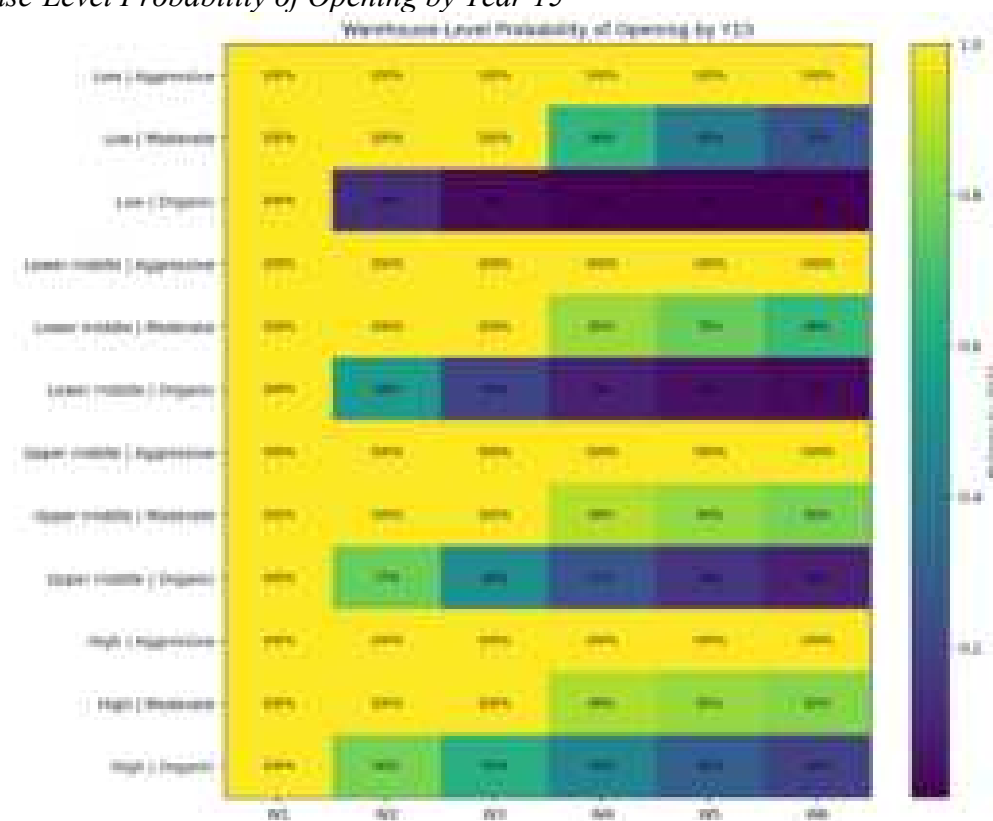
Applying the combined execution and financial definition changes the strategy ranking. Strict success is 1.3%, 1.5%, and 0.1% for Low-income Aggressive, Moderate, and Organic; 17.0%, 17.7%, and 3.1% in Lower-middle income; 37.1%, 38.7%, and 9.1% in Upper-middle income; and 69.4%, 70.6%, and 19.9% in High income.

Moderate has a small strict-success advantage over Aggressive in all four reference classes. The result does not make Moderate universally dominant. In High income the advantage is only about 1.2 percentage points, while Aggressive has the larger median NPV and greater mature scale. The result instead shows the value of conditional continuation: the first operating stage can screen later capital commitment.

Organic demonstrates why positive NPV cannot substitute for strategic completion. Its High-income positive-NPV frequency is the highest of the three strategies at 74.2%, but only 19.9% of paths complete the intended six-store network. Partial value creation is economically meaningful, but it is a different strategic outcome.

7.3 Warehouse Completion and Rollout Behavior

Figure 8
Warehouse-Level Probability of Opening by Year 15



Note. Aggressive completion is financing-insensitive by design; Moderate and Organic can experience financing-driven deferral in addition to construction delay.

Aggressive completes six warehouses in 100% of paths because Sponsor / External Capital supports the committed rollout. Construction can alter individual opening years, but financing does not prevent eventual completion within the horizon.

Moderate's first three sponsor-seeded warehouses are completed in essentially every path, while warehouses four through six increasingly depend on demonstrated funding capacity. Full completion is approximately 25.1% in Low income, 67.8% in Lower-middle income, 79.6% in Upper-middle income, and 81.9% in High income. Stronger operating states generate more cash and borrowing capacity, making the later stage easier to finance.

Organic produces the strongest completion constraint. Warehouse one opens because of the initial sponsor seed, but later warehouses depend on retained Project Cash. Full completion is only 0.1%, 3.1%, 9.2%, and 19.9% from Low through High income. The result is not a computational failure; it is the direct consequence of a financing policy that prioritizes capital preservation over guaranteed scale. Detailed execution and financing outcomes are reported in Appendix D, Exhibit D02.

7.4 Supporting Return Metrics

Table 5
Supporting Return and Payback Metrics Across the 12 Scenarios

Income group	Strategy	Median IRR	Median Effective WACC	Median MOIC	Median revenue CAGR	P(Payback by Y15)	Median payback year*
Low	Aggressive	-3.0%	20.6%	0.65x	10.3%	16.7%	13
Low	Moderate	-6.0%	20.6%	0.56x	12.1%	7.5%	14
Low	Organic	-0.6%	20.6%	0.87x	9.2%	19.5%	13
Lower-middle	Aggressive	6.5%	17.6%	1.91x	10.0%	57.5%	11
Lower-middle	Moderate	5.3%	17.6%	1.51x	13.6%	42.1%	12
Lower-middle	Organic	5.8%	17.6%	1.51x	11.4%	55.4%	11
Upper-middle	Aggressive	12.2%	15.3%	3.28x	10.1%	72.7%	9
Upper-middle	Moderate	12.4%	15.3%	3.10x	14.1%	65.6%	11
Upper-middle	Organic	11.7%	15.3%	2.07x	14.0%	65.4%	10
High	Aggressive	17.6%	10.4%	5.77x	11.3%	76.8%	8
High	Moderate	18.6%	10.4%	5.92x	15.7%	73.3%	10
High	Organic	19.1%	10.4%	3.59x	16.6%	76.8%	9

Note. *Median payback year is conditional on paths that achieve payback by Year 15. Effective WACC is the path-equivalent reporting hurdle, not the single rate used to discount every year.

The return measures reinforce the development gradient. Median IRR rises from negative values in Low income to roughly 17.6%-19.1% in High income, while median Effective WACC declines from approximately 20.6% to 10.4%. Country conditions therefore influence both the amount of operating cash flow and the return required to compensate capital providers.

High income is the only group in which median IRR clearly exceeds median Effective WACC under all three strategies. Upper-middle median IRRs remain below the 15.3% hurdle despite substantial positive upside, supporting the Yellow interpretation. High-income Moderate has the highest median MOIC at approximately 5.92x, while High-income Organic has the highest median IRR at approximately 19.1% but frequently remains below six-store scale.

Moderate also pays for conditionality through time: it has the slowest conditional median payback in every income group because later warehouses are deliberately delayed until operating evidence and financing capacity exist, leaving fewer years inside the 15-year window to recover their capital. Aggressive commits earlier and therefore gives successful stores more time to generate cash, while Organic can appear competitive on payback partly because it often invests much less and never reaches six-store scale. MOIC and revenue CAGR likewise answer different questions: MOIC measures the total multiple on deployed capital, whereas CAGR describes the growth rate of the realized revenue base. V5 therefore does not combine them into a single score, and a high CAGR in a staged or Organic path should not be confused with greater absolute value or scale.

7.5 Financing Exposure and the Cost of Execution

Table 6

Financing Exposure and Execution Outcomes

Income group	Strategy	Median cumulative Sponsor / External Capital (\$M)	Median peak Project Debt (\$M)	P(Financing delay)	Median cumulative financing-delay years	P(6/6 by Y15)
Low	Aggressive	318.3	0.0	0.0%	0	100.0%
Low	Moderate	159.4	7.1	100.0%	30	25.1%
Low	Organic	53.4	0.0	100.0%	71	0.1%
Lower-middle	Aggressive	286.8	0.0	0.0%	0	100.0%
Lower-middle	Moderate	145.6	24.3	96.9%	12	67.8%
Lower-middle	Organic	49.4	0.0	100.0%	66	3.1%
Upper-middle	Aggressive	286.8	0.0	0.0%	0	100.0%
Upper-middle	Moderate	146.6	31.9	89.1%	5	79.6%
Upper-middle	Organic	50.1	0.0	100.0%	58	9.2%
High	Aggressive	322.8	0.0	0.0%	0	100.0%
High	Moderate	165.5	55.8	74.4%	2	81.9%
High	Organic	57.0	0.0	100.0%	51	19.9%

Note. Cumulative financing-delay years sum warehouse-level deferrals and can exceed 15. Aggressive's zero Project Debt does not imply the sponsor itself uses no debt.

Aggressive requires the largest sponsor contribution, with median cumulative Sponsor / External Capital of roughly \$287-\$323 million across the groups. Moderate approximately halves the sponsor burden in many states and introduces Project Debt supported by the operating expansion. Median peak Project Debt rises from about \$7.1 million in Low income to \$55.8 million in High income because stronger projects can support more borrowing.

Organic requires only the initial sponsor seed, with median cumulative Sponsor / External Capital near \$49-\$57 million, but this capital defense is achieved by allowing later investment to wait when internally generated cash is insufficient. The strategy trade-off can therefore be summarized consistently: Aggressive pays for certainty with capital. Moderate pays for caution with time. Organic pays for capital preservation with scale.

7.6 Operating Economics and Terminal Cross-Check

The financial distributions ultimately originate in warehouse scale and destination-country operating economics. Gross profit is generated within each simulated pro forma as total project revenue less merchandise cost; membership revenue has no separate merchandise cost-of-goods-sold line in the project architecture. Gross profit is retained as an intermediate

operating output, while the Main Text emphasizes EBITDA, FCFF, and NPV because those measures provide the more direct bridge from operating economics to project valuation.

Aggressive produces the highest median Year-15 revenue, gross profit, EBITDA, FCFF, and terminal value in every income group. Complete Year-15 operating and terminal results, including gross profit, are reported in Appendix D, Exhibit D03.

In High income, median Year-15 revenue is approximately \$2.351 billion under Aggressive, \$2.251 billion under Moderate, and \$1.001 billion under Organic. Median EBITDA is approximately \$140.8 million, \$139.5 million, and \$62.3 million. Despite the scale difference, EBITDA margins are close - about 6.7%, 6.6%, and 6.6% - supporting the interpretation that strategy principally changes scale, timing, and financing while the country state primarily determines mature operating economics.

The Low-income cases show why mature operating profitability cannot be read as project value creation. Aggressive ends Year 15 with positive EBITDA of approximately \$19.6 million and positive FCFF of \$12.9 million, yet cumulative FCFF remains approximately -\$172.7 million; Moderate also ends with positive EBITDA and FCFF while cumulative FCFF remains about -\$140.1 million. The network can therefore look operationally viable late in the horizon without having recovered the capital absorbed during entry. By Upper-middle and High income, cumulative FCFF becomes strongly positive for Aggressive and Moderate, which helps explain why NPV improves much more sharply than terminal-year margins alone would suggest.

Terminal economics tell a related destination-versus-journey story. In High income, Aggressive and Moderate converge to similar mature EBITDA (approximately \$140.8 million and \$139.5 million) and terminal enterprise value (approximately \$1.038 billion and \$1.028 billion) even though they reach that mature state through different timing and financing paths.

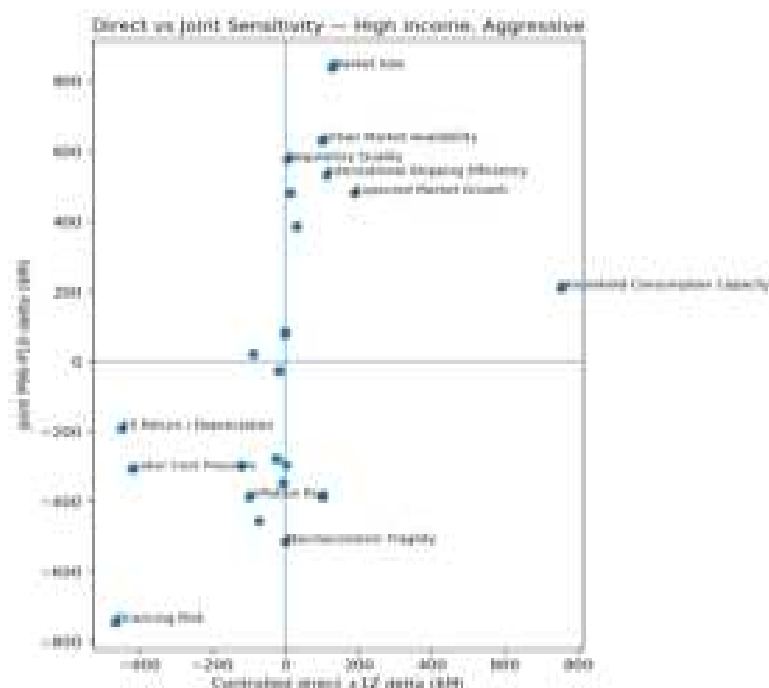
The resulting terminal EV/EBITDA is in the roughly 7x range and is used only as a DCF reasonableness cross-check, not as a competing valuation engine.

Annual P10, median, and P90 operating trajectories showing how these endpoint differences develop over the 15-year horizon are reported in Appendix D, Exhibit D04.

7.7 Direct Versus Joint Sensitivity

Sensitivity is evaluated in three ways because strategic importance, direct financial transmission, and correlated country-state association are not the same concept. Controlled direct sensitivity changes one factor while holding the broader country path controlled. Joint sensitivity asks what NPV outcomes tend to occur when high or low realizations of a factor appear as part of the complete correlated country system. A supplementary within-income-group scaling check then evaluates direct effects over each factor's actual P10-to-P90 range within the relevant reference class. Appendix C retains the two principal reporting views—controlled direct and joint correlated sensitivity—while the within-group scaling check is summarized in this section and retained in the reproducibility evidence.

Figure 9
Direct Versus Joint Country-Factor Sensitivity



Note. Direct bars represent controlled transmission; joint bars include the broader correlated country state. Joint differences are associations and must not be interpreted as single-factor causal coefficients.

Market Size illustrates the distinction. In High-income Aggressive, a controlled +/-1 Selected-Z intervention changes NPV by approximately \$126.5 million, while the joint P10-to-P90 NPV association is approximately \$839.7 million. The larger joint value does not represent an \$839.7 million Market Size coefficient; it reflects the broader favorable country configuration associated with market scale.

Macroeconomic Fragility provides the inverse example. Its controlled direct NPV coefficient is essentially zero because its final role is primarily through dependence and common Stress, yet its High-income Aggressive joint P10-to-P90 NPV association is approximately -\$515.1 million. Appendix C, Exhibits C01-C02 report the complete controlled and joint sensitivity evidence.

Across the full sensitivity exercise, the most consequential direct financial pathways cluster around demand scale and household consumption, financing conditions, and physical cost exposure such as labor and construction, while institutional and macro variables can become much more informative jointly because they travel with broader country states. A supplementary within-income-group scaling check also guards against mistaking a steep coefficient for a large practical driver when the factor barely varies inside the relevant class. For example, High-income Inflation Risk has only about a -\$12.2 million controlled P10-to-P90 NPV effect after the forward-inflation reconciliation. Strategic AHP weight, direct financial sensitivity, and joint association therefore answer different questions and should not be substituted for one another.

7.8 Integrated Interpretation

Research Question 2 is answered by the development gradient: income-group reference classes condition meaningfully different risk-return environments without being interpreted as causal country labels. Research Question 3 is answered by the financing trade-off: Aggressive maximizes execution and typical scale, Moderate conditions later capital on demonstrated financial capacity, and Organic protects sponsor capital while often accepting partial execution.

The development gradient should not be read as a simple 'high income equals low cost' relationship. High-income states can carry higher labor and construction costs, yet those disadvantages are often outweighed by stronger household consumption, institutions, logistics, financing conditions, and lower macroeconomic and FX risk. Lower-income states can offer cheaper physical inputs but still destroy value when demand capacity and required return move in the opposite direction. A central CST finding is therefore that attractiveness emerges from the joint configuration of Benefits, Costs, and Risks rather than from any single favorable input.

Research Question 4 does not produce one universal winner. High-income Aggressive is strongest on typical dollar value and mature scale. High-income Moderate is strongest on the combined strict full-project success criterion. High-income Organic can show strong capital-efficiency statistics while completing the intended network in only about one-fifth of paths. Management should therefore screen opportunity quality first and then select the financing policy that matches capital availability, conviction, urgency, and tolerance for incomplete execution.

8. Robustness, Validation, and Model Refinement

8.1 Structural Robustness

Robustness testing focuses on consequential assumptions that are not fully identified from direct data. The Moderate Project Debt thresholds, PPP reversion speed, terminal growth, company residual dependence, and network headroom were tested around their central specifications. The purpose is not to optimize a second model but to determine whether the broad decision zones and strategy conclusions depend on one fragile assumption.

Loosening Moderate credit constraints generally increases Project Debt usage and warehouse completion. In weak country states, however, more financing can slightly worsen median NPV because the project is enabled to invest more capital in poor economics. This reinforces the distinction between fundability and attractiveness: greater debt capacity cannot repair a value-destructive project.

PPP reversion and terminal growth affect valuation magnitude, especially in strong scenarios with larger continuation value, but the tested alternatives do not overturn the broad Low/Lower-middle/Upper-middle/High decision zones or the principal Aggressive-versus-Moderate trade-off. Network headroom sensitivity similarly changes individual path economics without dominating the six-store decision.

8.2 Computational and Economic Validation

Validation is performed at calculation, integration, economic, financing-feedback, and reproduction levels. The core architecture passes 75/75 tests; the final inflation/FX feature suite passes 22/22; financing-continuation invariants pass 60/60; and the V5 reproduction audit passes 12/12 scenarios. The reproduction errors are at floating-point scale, confirming that V5 adds the

IRR-versus-Effective-WACC reporting diagnostic without changing the previously frozen economic results. Detailed tests and tolerances are reported in Appendix A, Exhibit A05.

8.3 Model Refinement and Governance

Several important refinements were driven by economic diagnostics rather than software failure. A generic AHP-to-finance multiplier was replaced with factor-specific transmission. Independent country factors were replaced by empirical dependence. Current installed warehouse density was separated from economic network support. Inflation and FX were reconciled after a simpler formulation produced implausible long-horizon behavior. Financing evolved into a real execution feedback mechanism rather than a passive source-of-funds table.

Model governance also required distinguishing documentation corrections from economic recalibration. When an audit identified a difference between narrative documentation and the executed specification, the implementation was first verified against the underlying evidence and validation tests. Documentation was corrected when the numerical model was already valid, while economic recalibration was reserved for cases in which the executed behavior itself was not defensible.

The broader methodological lesson is that a Monte Carlo model can run perfectly and still be economically wrong. Defensible simulation requires source provenance, explicit distinctions between observed evidence and structural priors, deterministic reconstruction, economic plausibility checks, sensitivity analysis, and reproducible implementation.

9. Managerial Recommendations

9.1 Use Income Groups as Screening Gates, Not Final Decisions

World Bank reference classes are most useful as screening gates rather than final investment decisions. Low income is Red for the generic six-store experiment because even the upper valuation tail remains unfavorable, while Lower-middle income is Red / Selective because positive outcomes exist but are not typical. Upper-middle income is Yellow / Case-by-case because the negative median coexists with meaningful upside, making named-country evidence especially valuable. Management should therefore examine actual household purchasing power, metropolitan demand, site availability, permitting and construction conditions, supplier and logistics infrastructure, tax treatment, currency exposure, competitive capacity, and financing conditions before advancing an Upper-middle candidate. High income is Green for deeper underwriting, but meaningful downside remains and classification should never substitute for country-, metro-, and site-level due diligence.

9.2 Match Rollout Policy to Conviction and Capital Capacity

Aggressive is most defensible when conviction, sponsor capacity, and the strategic value of rapid scale are high. It produces the greatest typical value and mature scale in favorable environments but also the widest downside and largest capital exposure.

Moderate is the strongest balanced default in this experiment because it reduces sponsor exposure, preserves a route to full completion, and produces the highest strict full-project success frequency in every income class. Its advantage is conditionality rather than universal superiority; where delay is strategically costly, Aggressive can still be preferable.

Organic is appropriate when capital defense matters more than guaranteed scale. Its attractive percentage-return statistics must be read together with its low six-store completion rate because much of the protection comes from not committing later capital.

9.3 Recommended Decision Process

1. Screen the candidate set with company-specific Benefits, Costs, and Risks, using AHP or a comparable stakeholder process to make priorities explicit.

2. Translate material country conditions into defined operating, development, financing, timing, and valuation mechanisms.

3. Evaluate value, downside, return, payback, financing exposure, and completion under correlated uncertainty and alternative rollout policies.

4. Replace the income-group reference class with named-country evidence for candidates that survive screening.

5. Move from national evidence to region, metropolitan area, customer catchment, site, logistics, competition, and actual financing terms.

6. Make the investment decision and revisit continuation as operating evidence arrives.

9.4 Final Management Recommendation

For the defined six-warehouse experiment, management should prioritize High-income candidates for deeper due diligence, selectively investigate Upper-middle candidates with unusually favorable company-specific evidence, require exceptional justification for Lower-middle opportunities, and generally avoid advancing Low-income candidates under the generic specification. Moderate is the balanced starting policy, Aggressive fits high-conviction or rapid-scale cases, and Organic fits capital-defense priorities. First determine whether the opportunity deserves capital; then determine how much commitment the evidence justifies.

10. Limitations, Future Research, and Reflection

10.1 Decision and Generalizability Limitations

The most important limitation is geographic aggregation. V5 conditions on World Bank income groups rather than named countries. Countries within the same group can differ materially in purchasing power, cities, logistics, institutions, tax, construction, financing, and currency conditions. The results are therefore conditional reference-class distributions, not named-country forecasts.

Sponsor / External Capital is assumed available when the strategy calls for it. The model tracks how much outside project capital is required but does not estimate whether corporate management, banks, equity investors, or other capital providers would actually approve that funding. This matters especially when generalizing the architecture beyond a financially strong sponsor.

The AHP priorities are researcher-derived and Costco-informed rather than elicited from Costco executives or a multi-stakeholder panel. A low consistency ratio demonstrates internal coherence, not objective correctness or actual corporate preference.

10.2 Methodological and Calibration Limitations

The dependence matrix uses pairwise country samples from databases with different observation years. Pairwise coverage and bootstrap stability are audited, but the data are not a perfectly synchronized 23-variable panel. The Gaussian copula captures coherent rank dependence but may understate crisis-specific tail dependence, and one common dependence matrix is used across all four income groups.

The company residual history contains only ten annual Costco observations. V5 therefore uses bounded empirical residuals and does not pretend that a rich multivariate company time-

series model can be estimated from the available sample. The temporal country architecture also relies partly on transparent structural priors for persistence, stress transition, tax reform, and related dynamics.

Moderate Project Debt is an analytical rolling facility rather than a full loan contract. V5 omits collateral, fixed maturity, mandatory amortization, fees, covenant cure, refinancing, cross-default, and lender-specific underwriting. Actual transaction analysis would need to replace these abstractions with executable financing terms.

10.3 Scope Omissions

The experiment fixes the strategic scale at six warehouses and the explicit horizon at 15 years. It does not model abandonment, closure, relocation, sale, or acceleration beyond six stores. It does not optimize FX hedging, currency swaps, local debt as a natural hedge, repatriation policy, or local sourcing. Country-level network support also does not substitute for metropolitan catchments, competitor locations, travel times, parcel availability, or store-to-store overlap.

The model does not include an independent country variable for cultural acceptance of paid warehouse-club membership. Household consumption, urban concentration, membership productivity, and renewal capture related economics indirectly, but shopping habits, bulk-purchase preferences, vehicle access, storage capacity, and willingness to pay a membership fee could matter independently.

10.4 Future Research

The most immediate extension is named-country calibration. The World Bank income-group distribution could remain as a prior/reference class while the Year-0 state is replaced with actual candidate-country evidence. A second extension would move hierarchically from country

to region/state/province, metropolitan area, and site, allowing local demand, labor, regulation, logistics, real estate, competition, and construction economics to enter at the appropriate geographic level.

A stakeholder-panel AHP could replace the researcher-derived priority vector with separate judgments from finance, treasury, real estate, operations, merchandising, logistics, legal/regulatory, senior management, and external capital providers. A richer fundability module could then model sponsor approval, actual loan structures, dilution, investor return requirements, covenant structures, and capital availability.

Longer country panels could support income-group-specific persistence, regime models, volatility clustering, and alternative copulas with stronger tail dependence. A treasury extension could compare unhedged exposure with forwards, swaps, local-currency debt, and natural hedges. A successor model could also make expansion, waiting, closure, or re-entry fully endogenous rather than beginning with a predetermined six-store plan.

The architecture is also portable across industries, but the calibration is not. A different business model would require a different factor set, stakeholder priorities, financial-transmission rules, company residuals, and scale assumptions while retaining the broader sequence from screening to financial transmission, financing/execution policy, and valuation under uncertainty.

10.5 Research and Methodological Reflection

A central lesson from the project is that technical execution and economic validity are not the same thing. A Monte Carlo model can run without errors, produce polished probability distributions, and still be economically wrong. The FX reconciliation, network-support recalibration, warehouse-ramp lineage audit, construction-delay documentation audit, and financing terminology review each demonstrated a different form of this risk.

The correct response was not to manufacture precision. Unsupported bankruptcy thresholds were not added merely to produce a survival probability. A six-dimensional company copula was not fitted to ten annual observations. The AHP weights were not converted into financial coefficients simply because they were numeric. Literature was used to support mechanisms and methods without implying that outside authors estimated Costco-specific transmission coefficients.

Validation therefore became part of model development rather than a final software check. The research process increasingly followed a governance sequence of source provenance, explicit classification of observed versus derived versus assumed inputs, deterministic reconstruction, economic plausibility testing, stochastic validation, sensitivity analysis, and revision when the implied economics were not defensible.

The final model is consequently a proof of concept in model governance as much as Monte Carlo valuation. Its value is not that uncertainty disappears. Its value is that assumptions, mechanisms, dependencies, execution constraints, and decision boundaries become sufficiently explicit that a reader can challenge or replace them without rebuilding the entire framework.

10.6 AI Use Note

Generative artificial intelligence, primarily OpenAI's ChatGPT, was used as a research and technical support tool during this independent study, including for brainstorming, code development and debugging, model review, documentation, and editing. AI outputs were not treated as research evidence, and final source selection, modeling decisions, interpretation, and submitted work remained the responsibility of the author. A more detailed disclosure of AI use and research safeguards is provided in Appendix F.

11. Conclusion

This study develops a bridge between company-specific country screening and financial valuation under uncertainty. CST identifies the external conditions that matter, AHP makes strategic priorities explicit, the financial-transmission layer maps country states into operating and valuation mechanics, and Monte Carlo simulation evaluates the resulting distributions. Costco provides the warehouse-club proof of concept, but the contribution is the modular architecture rather than a universal Costco calibration.

The experiment shows that World Bank income groups can serve as useful empirical reference classes without becoming deterministic country labels. Low and Lower-middle income are unfavorable under the generic six-store specification, Upper-middle income is a transition zone with meaningful upside and high value of additional information, and High income is the strongest screening environment. Strategy then changes how that opportunity is pursued: Aggressive buys execution certainty with capital, Moderate buys conditionality with time, and Organic protects capital at the cost of scale.

The final decision rule is sequential: determine whether the opportunity deserves capital, then determine how much commitment the evidence justifies. A reusable international-expansion model can therefore connect strategic screening, country dependence, business-model economics, financing constraints, execution policy, and valuation without collapsing them into one opaque score. Named-country, metropolitan, site, treasury, and transaction underwriting remain necessary before a real investment decision.

References

- Andersen, T. G., & Bollerslev, T. (1996). *Deutsche Mark-Dollar volatility: Intraday activity patterns, macroeconomic announcements, and longer run dependencies* (NBER Working Paper No. 5783). National Bureau of Economic Research. <https://doi.org/10.3386/w5783>
- Anton, S. G., & Afloarei Nucu, A. E. (2022). On the role of institutional factors in shaping working capital management policies: Empirical evidence from European listed firms. *Economic Systems*, 46(2), 100976. <https://doi.org/10.1016/j.ecosys.2022.100976>
- Arvis, J.-F., Herrera Dappe, M., Ulybina, D., & Wiederer, C. (2026). *The new Logistics Performance Indicators 2.0 (LPI 2.0): Methodology and user guide*. World Bank Group. <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099042226142027181>
- Bank for International Settlements. (n.d.). *Bilateral exchange rates* [Data set]. BIS Data Portal. Retrieved August 21, 2026, from <https://data.bis.org/topics/XRU>
- Board of Governors of the Federal Reserve System. (n.d.). *Nominal yield curve*. Retrieved August 21, 2026, from <https://www.federalreserve.gov/data/nominal-yield-curve.htm>
- Borio, C., Lombardi, M., Yetman, J., & Zakrajšek, E. (2023). *The two-regime view of inflation* (BIS Papers No. 133). Bank for International Settlements. <https://www.bis.org/publ/bppdf/bispap133.htm>
- Brandão-Marques, L., Meeks, R., & Nguyen, V. (2024). *Monetary policy with uncertain inflation persistence* (IMF Working Paper No. 2024/047). International Monetary Fund. <https://doi.org/10.5089/9798400269394.001>
- Cavusgil, S. T., Kiyak, T., & Yeniyurt, S. (2004). Complementary approaches to preliminary foreign market opportunity assessment: Country clustering and country ranking.

Industrial Marketing Management, 33(7), 607–617.

<https://doi.org/10.1016/j.indmarman.2003.10.005>

Costco Wholesale Corporation. (2016). *Annual report on Form 10-K for the fiscal year ended August 28, 2016*. U.S. Securities and Exchange Commission.

<https://www.sec.gov/Archives/edgar/data/909832/000090983216000032/cost10k82816.htm>

Costco Wholesale Corporation. (2017). *Annual report on Form 10-K for the fiscal year ended September 3, 2017*. U.S. Securities and Exchange Commission.

<https://www.sec.gov/Archives/edgar/data/909832/000090983217000014/cost10k90317.htm>

Costco Wholesale Corporation. (2018). *Annual report on Form 10-K for the fiscal year ended September 2, 2018*. U.S. Securities and Exchange Commission.

<https://www.sec.gov/Archives/edgar/data/909832/000090983218000013/cost10k9218.htm>

Costco Wholesale Corporation. (2019). *Annual report on Form 10-K for the fiscal year ended September 1, 2019*. U.S. Securities and Exchange Commission.

<https://www.sec.gov/Archives/edgar/data/909832/000090983219000019/cost10k9119.htm>

Costco Wholesale Corporation. (2020). *Annual report on Form 10-K for the fiscal year ended August 30, 2020*. U.S. Securities and Exchange Commission.

<https://www.sec.gov/Archives/edgar/data/909832/000090983220000017/cost-20200830.htm>

Costco Wholesale Corporation. (2021). *Annual report on Form 10-K for the fiscal year ended August 29, 2021*. U.S. Securities and Exchange Commission.

<https://www.sec.gov/Archives/edgar/data/909832/000090983221000014/cost-20210829.htm>

Costco Wholesale Corporation. (2022). *Annual report on Form 10-K for the fiscal year ended August 28, 2022*. U.S. Securities and Exchange Commission.

<https://www.sec.gov/Archives/edgar/data/909832/000090983222000021/cost-20220828.htm>

Costco Wholesale Corporation. (2023). *Annual report on Form 10-K for the fiscal year ended September 3, 2023*. U.S. Securities and Exchange Commission.

<https://www.sec.gov/Archives/edgar/data/909832/000090983223000042/cost-20230903.htm>

Costco Wholesale Corporation. (2024). *Annual report on Form 10-K for the fiscal year ended September 1, 2024*. U.S. Securities and Exchange Commission.

<https://www.sec.gov/Archives/edgar/data/909832/000090983224000049/cost-20240901.htm>

Costco Wholesale Corporation. (2025). *Annual report on Form 10-K for the fiscal year ended August 31, 2025*. U.S. Securities and Exchange Commission.

<https://www.sec.gov/Archives/edgar/data/909832/000090983225000101/cost-20250831.htm>

Damodaran, A. (2026a). *Corporate marginal tax rates - By country* [Data set]. NYU Stern School of Business.

https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/countrytaxrates.html

Damodaran, A. (2026b). *Country default spreads and risk premiums* [Data set]. NYU Stern School of Business.

https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/ctryprem.html

Damodaran, A. (2026c). *Historical implied equity risk premiums* [Data set]. NYU Stern School of Business.

https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histimpl.html

Einav, L., Klenow, P. J., Levin, J. D., & Murciano-Goroff, R. (2021). *Customers and retail growth* (NBER Working Paper No. 29561). National Bureau of Economic Research.

<https://doi.org/10.3386/w29561>

Federal Open Market Committee. (2026, January 28). *Federal Open Market Committee reaffirms its “Statement on Longer-Run Goals and Monetary Policy Strategy.”* Board of Governors of the Federal Reserve System.

<https://www.federalreserve.gov/newsevents/pressreleases/monetary20260128b.htm>

Frankel, J. A., & Rose, A. K. (1995). *A panel project on purchasing power parity: Mean reversion within and between countries* (NBER Working Paper No. 5006). National Bureau of Economic Research. <https://doi.org/10.3386/w5006>

Fund for Peace. (n.d.). *Fragile States Index methodology*.

<https://fragilestatesindex.org/methodology/>

Gilchrist, S., Wei, B., Yue, V. Z., & Zakrajšek, E. (2021). *Sovereign risk and financial risk* (NBER Working Paper No. 29501). National Bureau of Economic Research.

<https://doi.org/10.3386/w29501>

- Gompers, P. A. (1995). Optimal investment, monitoring, and the staging of venture capital. *The Journal of Finance*, 50(5), 1461–1489. <https://doi.org/10.1111/j.1540-6261.1995.tb05185.x>
- Guler, A. U. (2018). Inferring the economics of store density from closures: The Starbucks case. *Marketing Science*, 37(4), 611–630. <https://doi.org/10.1287/mksc.2017.1078>
- Gürkaynak, R. S., Sack, B., & Wright, J. H. (2006). *The U.S. Treasury yield curve: 1961 to the present* (Finance and Economics Discussion Series 2006-28). Board of Governors of the Federal Reserve System.
<https://www.federalreserve.gov/pubs/feds/2006/200628/200628abs.html>
- Hacura, A., Jadamus-Hacura, M., & Kocot, A. (2001). Risk analysis in investment appraisal based on the Monte Carlo simulation technique. *The European Physical Journal B*, 20(4), 551–553. <https://doi.org/10.1007/s100510170238>
- Hamilton Lane. (n.d.). *J-curves: An introduction*. <https://www.hamiltonlane.com/en-us/knowledge-center/j-curves>
- Higham, N. J. (2002). Computing the nearest correlation matrix - A problem from finance. *IMA Journal of Numerical Analysis*, 22(3), 329–343. <https://doi.org/10.1093/imanum/22.3.329>
- Holmes, T. J. (2008). *The diffusion of Wal-Mart and economies of density* (NBER Working Paper No. 13783). National Bureau of Economic Research.
<https://doi.org/10.3386/w13783>
- ICE Data Indices, LLC. (n.d.). *ICE BofA US Corporate Index Option-Adjusted Spread [BAMLC0A0CM]* [Data set]. Federal Reserve Bank of St. Louis, FRED. Retrieved August 21, 2026, from <https://fred.stlouisfed.org/series/BAMLC0A0CM>

- International Labour Organization. (n.d.-a). *Statistics on unemployment and labour underutilization*. ILOSTAT. Retrieved August 21, 2026, from <https://ilostat.ilo.org/topics/unemployment-and-labour-underutilization/>
- International Labour Organization. (n.d.-b). *Wages and working time statistics (COND database)*. ILOSTAT. Retrieved August 21, 2026, from <https://ilostat.ilo.org/methods/concepts-and-definitions/description-wages-and-working-time-statistics/>
- International Monetary Fund. (2026). *World Economic Outlook: Global economy in the shadow of war*. <https://www.imf.org/en/publications/weo/issues/2026/04/14/world-economic-outlook-april-2026>
- Kim, Y., & Choi, W. G. (2003). *Trade credit and the effect of macro-financial shocks: Evidence from U.S. panel data* (IMF Working Paper No. 03/127). International Monetary Fund. <https://doi.org/10.5089/9781451855005.001>
- Knight, F. H. (1921). *Risk, uncertainty and profit*. Houghton Mifflin. <https://fraser.stlouisfed.org/title/risk-uncertainty-profit-110>
- Levary, R. R., & Wan, K. (1999). An analytic hierarchy process based simulation model for entry mode decision regarding foreign direct investment. *Omega*, 27(6), 661–677. [https://doi.org/10.1016/S0305-0483\(99\)00032-8](https://doi.org/10.1016/S0305-0483(99)00032-8)
- Lim, S. F. W. T., Rabinovich, E., Park, S., & Hwang, M. (2021). Shopping activity at warehouse club stores and its competitive and network density implications. *Production and Operations Management*, 30(1), 28–46. <https://doi.org/10.1111/poms.13226>
- McDonald, R. L., & Siegel, D. (1986). The value of waiting to invest. *The Quarterly Journal of Economics*, 101(4), 707–727. <https://doi.org/10.2307/1884175>

Nini, G., Smith, D. C., & Sufi, A. (2009). Creditor control rights and firm investment policy.

Journal of Financial Economics, 92(3), 400–420.

<https://doi.org/10.1016/j.jfineco.2008.04.008>

S&P Global Ratings. (2024, January 7). *Corporate methodology*.

<https://www.spglobal.com/ratings/en/regulatory/article/-/view/sourceId/12913251>

Saaty, T. L. (2008). Decision making with the analytic hierarchy process. *International Journal*

of Services Sciences, 1(1), 83–98. <https://doi.org/10.1504/IJSSCI.2008.017590>

Savvides, S. (1994). Risk analysis in investment appraisal. *Project Appraisal*, 9(1), 3–18.

<https://doi.org/10.1080/02688867.1994.9726923>

Sufi, A. (2009). Bank lines of credit in corporate finance: An empirical analysis. *The Review of*

Financial Studies, 22(3), 1057–1088. <https://doi.org/10.1093/revfin/hhm007>

Transparency International. (2026, February 10). *Corruption Perceptions Index 2025*.

<https://www.transparency.org/en/cpi/2025>

World Bank. (2025a). *International Comparison Program 2021 methodology*.

[https://thedocs.worldbank.org/en/doc/c74e68c6b6a87095e14da36f31f99c5b-](https://thedocs.worldbank.org/en/doc/c74e68c6b6a87095e14da36f31f99c5b-0050022025/icp-2021-methodology)

[0050022025/icp-2021-methodology](https://thedocs.worldbank.org/en/doc/c74e68c6b6a87095e14da36f31f99c5b-0050022025/icp-2021-methodology)

World Bank. (2025b). *Worldwide Governance Indicators* [Data set] (2025 Revision). Retrieved

August 21, 2026, from [https://www.worldbank.org/en/publication/worldwide-](https://www.worldbank.org/en/publication/worldwide-governance-indicators)

[governance-indicators](https://www.worldbank.org/en/publication/worldwide-governance-indicators)

World Bank. (2026a). *B-READY methodology handbook 2026*. World Bank Group.

[https://thedocs.worldbank.org/en/doc/6364c306d685203c859c60a075df5c3a-](https://thedocs.worldbank.org/en/doc/6364c306d685203c859c60a075df5c3a-0540012026/original/B-READY-MH-2026.pdf)

[0540012026/original/B-READY-MH-2026.pdf](https://thedocs.worldbank.org/en/doc/6364c306d685203c859c60a075df5c3a-0540012026/original/B-READY-MH-2026.pdf)

World Bank. (2026b). *Financial Management Information Systems database* [Data set] (July 2026 version). Retrieved August 21, 2026, from

<https://datacatalog.worldbank.org/search/dataset/0037882/financial-management-information-systems-database>

World Bank. (2026c). *World Development Indicators* [Data set]. Retrieved August 21, 2026, from <https://datacatalog.worldbank.org/search/dataset/0037712/world-development-indicators>

World Bank. (n.d.). *World Bank country and lending groups*. Retrieved August 21, 2026, from <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups>

Appendix A - Model Architecture and Reproducibility

Appendix A provides the mathematical and calibration detail required to reproduce the final V5 financial architecture. Unless otherwise stated, equations and values correspond to the frozen executable. External literature supports data, methods, or economic mechanisms; it should not be interpreted as estimating project-specific coefficients unless explicitly noted.

Exhibit A01 - Master Equation Registry

Module	Executed equation / rule	Interpretation
Income-group marginal	$U_i = \Phi(Y_i); Z_i = F^{-1}_{\{g,i\}}(U_i)$	Maps correlated latent percentile into the group-specific empirical marginal.
Year-0 dependence	$Y_0 \sim N(0, R)$	R is the final PSD-repaired 23x23 latent Gaussian matrix.
Anchored temporal state	$a_i = \sqrt{\rho_0} Y_{i0}; H_{it} = a_i + \phi_i(H_{i,t-1} - a_i) + \text{innovation}$	Preserves country identity while allowing annual evolution.
CPI	$CPI_t = CPI_{t-1}(1 + \pi_t)$	Local nominal price index.
Tax reform	With $p=5\%/yr$, statutory tax changes $\pm 1/2/3$ pp	Sticky policy state.
Stress vulnerability	$v_t = .5$ adverse Political Stability percentile + $.5$ adverse Macro Fragility percentile	Conditions entry into the Stress regime.
Stress entry	$p_{NS} = 0.025 + 0.05 v_t$	Approximately 2.5%-7.5% annual entry.
Default spread	$DS_t = \max(0, 3.7074556\% + 3.442058 \text{ pp} * Z_{21})$	Factor-21 reconstruction.
Country risk premium	$CRP_t = 1.523 DS_t$	Central country-risk scaling.
Expected FX depreciation	$m_t = \text{clip}(5.9359\% + 23.4372 \text{ pp} * Z_{22})$	Factor-22 reconstruction.
FX volatility	$\sigma_{FX,t} = \text{clip}(\max(0, 8.9108\% + 13.3548 \text{ pp} * Z_{23}))$	Factor-23 reconstruction.
FX return	$\Delta \ln FX_t = \text{central depreciation} + .15 \ln(q_{t-1}) + \sigma_{FX,t} \epsilon_{FX,t}$	Inflation/PPP-consistent realized currency process.
Delay incidence	$q = \text{logistic}(-1.38629 + 0.40547 Z_{20})$	Probability of positive construction delay.
Two-year conditional delay	$s = \text{logistic}(-1.38629 + 0.22314 Z_{20})$	Probability delay=2 conditional on positive delay.
Warehouse age	$\text{Age}_{wt} = t - \text{OpeningYear}_w + 1$	Selects audited 10-year age-productivity benchmark.
Demand access	$DA_t = .5 Z_{4,t} + .5 Z_{5,t}$	Combines metro concentration and urban availability.
Network support	$S_0 = 5.3598 \exp(0.80 \sigma_{MS} Z_{1,0} + 0.30 \sigma_{HC} Z_{2,0})$	Economic support capacity.
Post-support incrementality	$m(U)=1$ for $U \leq 1$; otherwise $1/[1+(1/.75-1)(U-1)^2]$	75% marginal productivity at $U=2$.
EBITDA	Revenue - COGS - Cash SG&A - Preopening	Operating profit before D&A.
Operating NWC	Inventory - Accounts Payable	Modeled operating working capital.
FCFF	EBIT - Cash Tax + D&A - Delta NWC - Construction CapEx - Maintenance CapEx	Unlevered free cash flow.
Cost of equity	$K_{e,t} = R_{f,t} + .91(4.23\%) + (CRP_t - 0.23\%)$	Exact frozen V5 equation; 0.23% is frozen U.S. reference constant.
Parent/proxy cost of debt	$K_{d,t} = R_{f,t} + 0.79\%$	Strong-parent debt proxy.
WACC	$0.987063 K_{e,t} + 0.012937 K_{d,t}(1 - \text{tax}_t)$	Enterprise valuation hurdle.
Discount factor	$DF_t = DF_{t-1}/(1 + WACC_t)$	Annual path-specific discounting.
Terminal growth	$g^* = \min(2.5\%, \max(0, WACC_{\text{bar}} - 1.0\%))$	Continuation-value guardrail.
Terminal value	$TV_{15} = FCFF_{16}/(WACC_{\text{bar}} - g^*)$	Continuation value of the realized operating network beyond Year 15.
NPV	$\sum FCFF_t DF_t + TV_{15} DF_{15}$	Sponsor-currency enterprise NPV.
Effective WACC	$1 + WACC_{\text{eff}} = [\text{product}_t(1 + WACC_t)]^{(1/15)}$	Path-equivalent reporting hurdle.
Moderate debt capacity	$\min(3.0x \text{ EBITDA}, \text{EBITDA}/(6.0x K_{d,\text{proj}}), \text{maintenance cash flow}/15\%)$	Most restrictive borrowing-capacity test governs.
Strict full success	6/6 by Y15 AND NPV > 0 AND IRR > Effective WACC	Final combined strategic/financial criterion.

Note. The frozen V5 cost-of-equity equation subtracts a 0.23% U.S. reference constant from the scaled destination-country risk premium. The term is reproduced exactly as executed and is not interpreted as a separately estimated U.S. country-risk premium.

Exhibit A02 - Assumption and Calibration Registry

Module / parameter	Final V5 value or method	Classification / basis
Horizon / scale	15 years; maximum 6 warehouses	Structural research design
Paths / seed	10,000 per scenario; seed 499	Computational design
Conditioning groups	Low, Lower-middle, Upper-middle, High income	World Bank reference classes
Stochastic variables	23	21 CST factors + expected FX depreciation + FX volatility
Marginal grid	6,000 points per factor/group	Derived empirical calibration
Marginal diagnostic draws	100,000 per factor/group	Validation
Joint dependence validation	50,000 draws per income group	Validation
Deep structural persistence	$\rho=.90, \phi=.90$	Structural temporal prior
Cyclical persistence	$\rho=.40, \phi=.60$	Structural temporal prior
Risk/Stress persistence	$\rho=.65, \phi=.75$	Structural temporal prior
Inflation persistence	$\rho=.30, \phi=.70$	Structural temporal prior
Construction-delay state	$\rho=.75, \phi=.85$	Structural temporal prior
Expected FX depreciation	$\rho=.20, \phi=.35$	Structural temporal prior
FX volatility	$\rho=.50, \phi=.75$	Structural temporal prior
Stress entry / persistence	2.5%-7.5% entry; 50% persistence	Structural prior
Forward inflation anchors	4.0%, 3.5%, 3.0%, 2.0%	Literature-informed structural priors
PPP correction	15% per year	Literature-informed structural prior
Mature merchandise revenue	\$111.979M / warehouse	Derived True-Zero company calibration
Mature membership revenue	\$1.734M / warehouse	Derived True-Zero company calibration
Warehouse age curve	10-year deterministic cohort/calendar benchmark	Costco FY2025 derived calibration
Separate ramp residual	None	Scope / evidence boundary
Base MCR	90.0091%	Derived True-Zero calibration
Base cash SG&A	3.7214%	Derived True-Zero calibration
Inventory / AP days	34.445 / 31.906	Derived True-Zero calibration
Initial network support	5.3598 warehouse equivalents	3.5732 legacy density x 1.50 headroom
Post-support anchor	75% productivity at U=2	Structural prior
Maintenance CapEx	1.0% of merchandise revenue	Structural calibration
ERP / beta	4.23% / 0.91	Market/company calibration
Corporate debt spread	0.79%	Current market calibration
Parent/proxy weights	98.7063% equity / 1.2937% debt	Derived company calibration
Terminal growth	2.5% with 1 pp WACC-g floor	Structural valuation prior
Moderate thresholds	3.0x leverage; 6.0x interest coverage; 15% maintenance cash flow/Debt	Researcher-defined credit policy
Aggressive planned openings	1/1/1/1/1	Strategy design
Moderate planned openings	1/1/1/5/5/5	Strategy design
Organic planned openings	1/2/3/4/5/6	Strategy design

Note. Structural and literature-informed priors are explicitly identified where direct project-specific estimation was not supported by the available data. A citation beside a prior supports the mechanism or context, not necessarily the exact coefficient.

Exhibit A03 - True-Zero Numerical Transition and U.S. Reconstruction

Variable	U.S. target	True Zero	U.S. reconstructed	Validation
Mature merchandise revenue (\$M)	287.000	111.979	287.000	PASS
Mature membership revenue (\$M)	5.90133	1.734	5.90173	PASS
Real warehouse growth	4.03645%	4.18224%	4.03645%	PASS
Merchandise-cost ratio	88.87563%	90.00910%	88.87563%	PASS
Cash SG&A ratio	8.35087%	3.72140%	8.35089%	PASS
Inventory days	28.9208	34.4450	28.9205	PASS
Accounts-payable days	30.6660	31.9060	30.6665	PASS
Construction CapEx (\$M)	75.000	49.877	75.346	PASS*
Preopening (\$M)	3.000	1.522	2.99968	PASS
Effective tax rate	25.13404%	21.94650%	25.13404%	PASS
Default spread	0.23000%	3.70746%	0.23000%	PASS

Note. *Construction CapEx uses the pre-specified wider absolute tolerance because soft-development friction is modeled separately. True Zero is the global statistical center, not an actual neutral country.

Exhibit A04 - Deterministic Financing-Aware Strategy Pro Formas

Metric	Aggressive	Moderate	Organic
Planned opening years	1/1/1/1/1/1	1/1/1/5/5/5	1/2/3/4/5/6
Financing-adjusted openings	1/1/1/1/1/1	1/1/1/5/6/7	1/10/14/16/17/18
Warehouses operating by Y15	6	6	3
Cumulative Sponsor / External Capital (\$M)	308.394	154.197	51.399
Peak Project Debt (\$M)	0.000	50.008	0.000
NPV (\$M)	-9.852	+2.033	-7.337
IRR	13.67%	14.36%	12.54%
Effective WACC	14.34%	14.34%	14.34%
MOIC	3.80x	4.11x	2.30x
Payback	Year 9	Year 11	No payback by Y15
Y15 revenue (\$M)	1,209.847	1,202.847	505.556
Y15 EBITDA (\$M)	93.144	92.604	37.400
Y15 FCFE (\$M)	62.967	63.184	-24.207
Terminal value (\$M)	512.880	509.905	205.965

Note. This deterministic True-Zero audit isolates strategy mechanics with stochastic country/company variation disabled.

Exhibit A05 - Validation and Reproduction Audit

Validation suite	Tests passed	Status
Core architecture / numerical validation	75 / 75	PASS
V4.1 inflation/FX feature validation	22 / 22	PASS
V4.2 financing-continuation invariants	60 / 60	PASS
V5 12-scenario reproduction	12 / 12	PASS

The core suite tests input dimensions, PSD dependence, deterministic reconstruction, U.S. inverse reconstruction, latent temporal behavior, FX/CPI guardrails, Stress incidence, construction timing, network mechanics, finite valuation, and terminal-value guardrails. The V4.1 feature suite validates the reconciled inflation/FX architecture and controlled sensitivity identities. The V4.2 suite tests 15 financing invariants in each income group. V5 then reproduces all 12 scenario NPV and IRR arrays to floating-point precision.

The overall maximum absolute V5 reproduction error is approximately $9.095\text{e-}13$ million dollars for NPV and $4.441\text{e-}16$ for IRR. These are numerical floating-point differences rather than economic changes. Validation counts are implementation checks, not statistical confidence levels.

Exhibit A06 - 10-Year Warehouse-Age Productivity Calibration

Warehouse age	Costco FY2025 cohort evidence (\$M)	Relative mature productivity	True-Zero merchandise (\$M)	True-Zero membership (\$M)
1	192	66.90%	74.913	1.160
2	192	66.90%	74.913	1.160
3	186	64.81%	72.572	1.124
4	201	70.03%	78.424	1.214
5	210	73.17%	81.936	1.269
6	240	83.62%	93.641	1.450
7	242	84.32%	94.421	1.462
8	245	85.37%	95.592	1.480
9	277	96.52%	108.077	1.673
10+	287	100.00%	111.979	1.734

Note. The curve is an empirical FY2025 cohort/calendar benchmark, not a causal universal maturation law. V5 preserves the non-monotonic third observation and samples no independent stochastic warehouse-ramp residual.

Appendix B - CST and AHP Evidence

Appendix B documents the strategic-governance and country-state input systems. The 21x21 AHP matrix represents researcher-derived strategic importance; the 23x23 stochastic matrix represents empirically calibrated dependence. They answer different questions and are not interchangeable.

Exhibit BAP01 - Full 21x21 AHP Pairwise Comparison Matrix

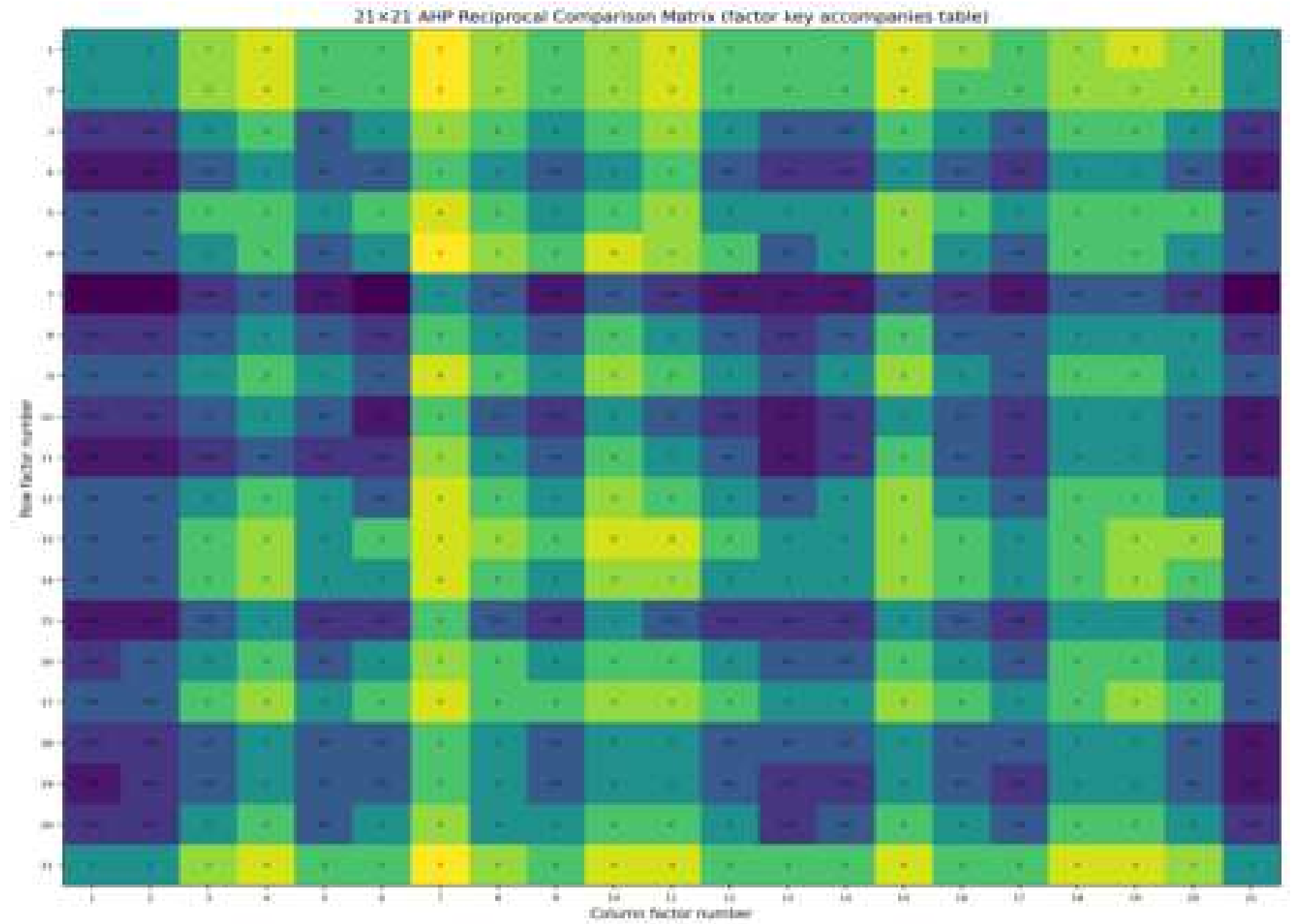


Exhibit BAP02 - AHP Consistency Diagnostics

Diagnostic	Final result
Criteria, n	21
Unique pairwise comparisons	210
Completed comparisons	210
lambda max	21.5356878
Consistency Index (CI)	0.0267844
Random Index (RI)	1.64
Consistency Ratio (CR)	0.0163319
Consistency Ratio (%)	1.633%
Acceptance threshold	10.000%
Assessment	Acceptable
Final weight total	1.000000

The low CR demonstrates internal consistency of the researcher-derived judgment system; it does not establish that the priorities are objectively correct, that Costco management would choose identical weights, or that the weights are statistically estimated financial coefficients.

Exhibit BAP03 - Complete 23-Factor Empirical Marginal Distributions

Each stochastic variable uses an income-group-specific empirical marginal distribution on the common Selected Global-Z scale. The simulation samples the full 6,000-point numerical grids; the figures below display the final calibrated density evidence. Selected raw-versus-ln transformations are documented in Appendix E, Exhibit E01.

Figure B1. Market Size

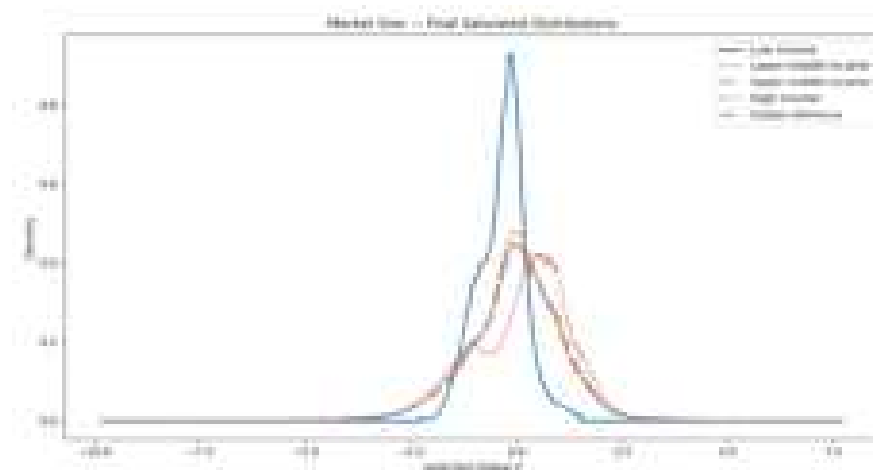


Figure B2. Household Consumption Capacity

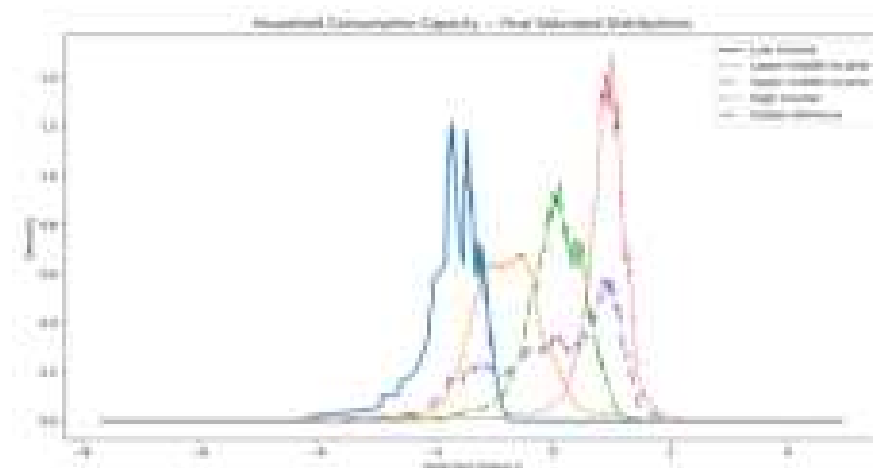


Figure B3. Expected Market Growth

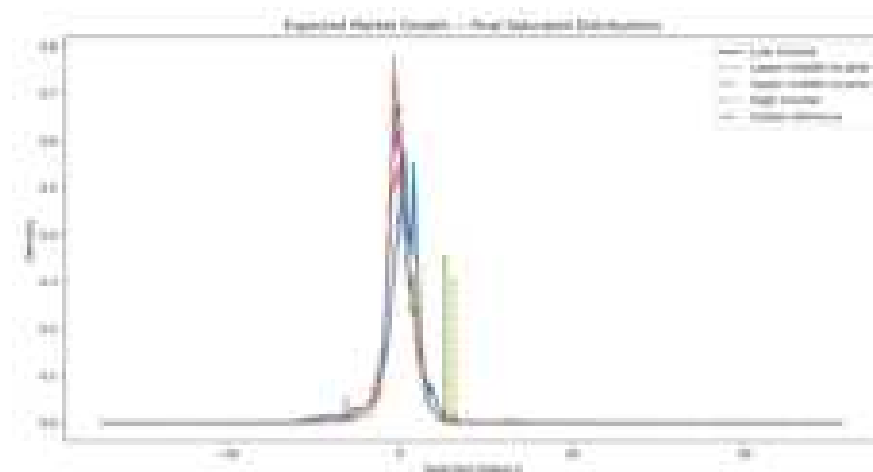


Figure B4. Metropolitan Customer Concentration

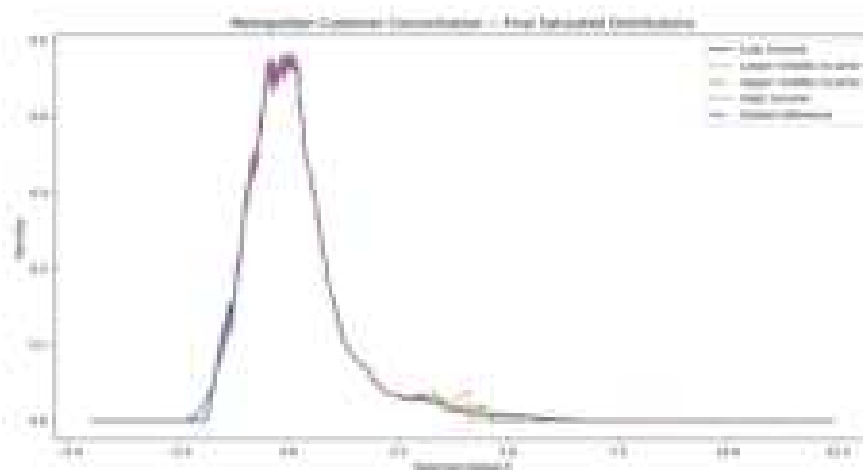


Figure B5. Urban Market Availability

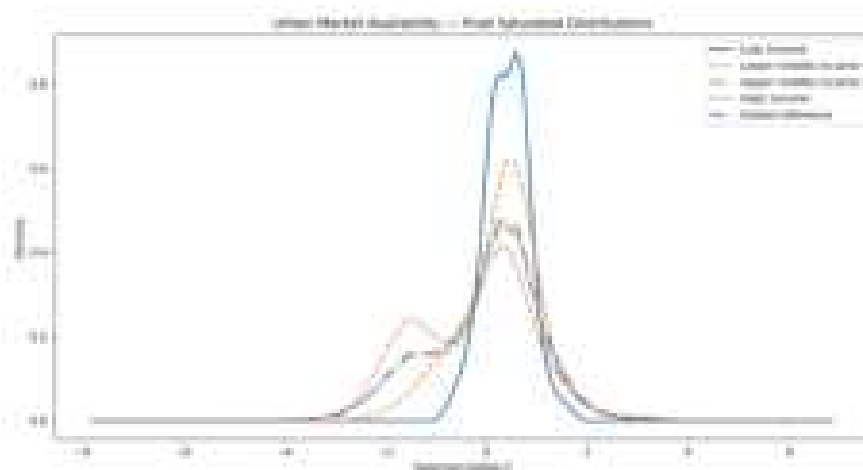


Figure B6. Labor Cost Pressure

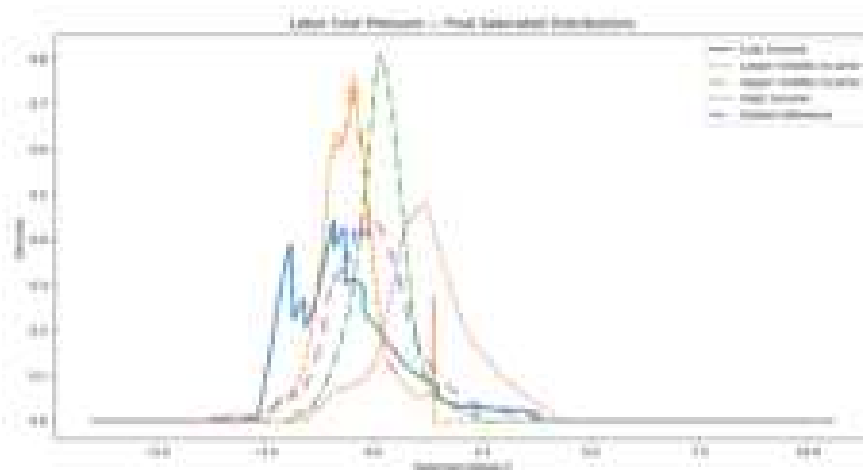


Figure B7. Labor Underutilization

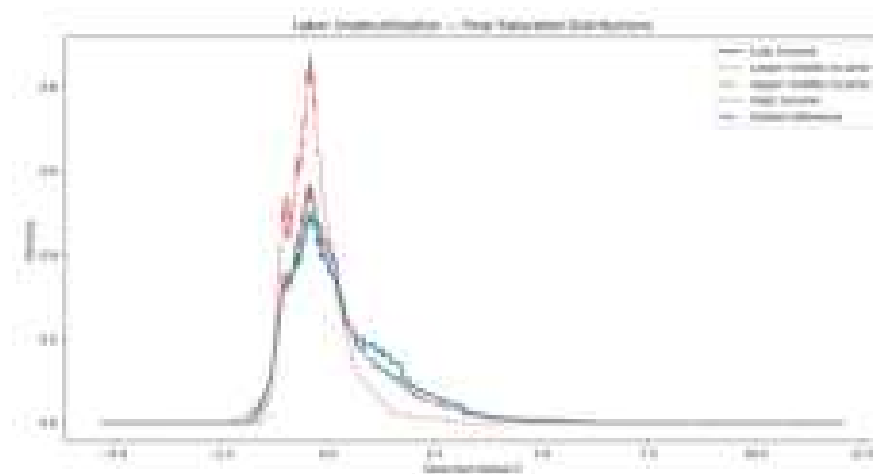


Figure B8. Site Access Pressure

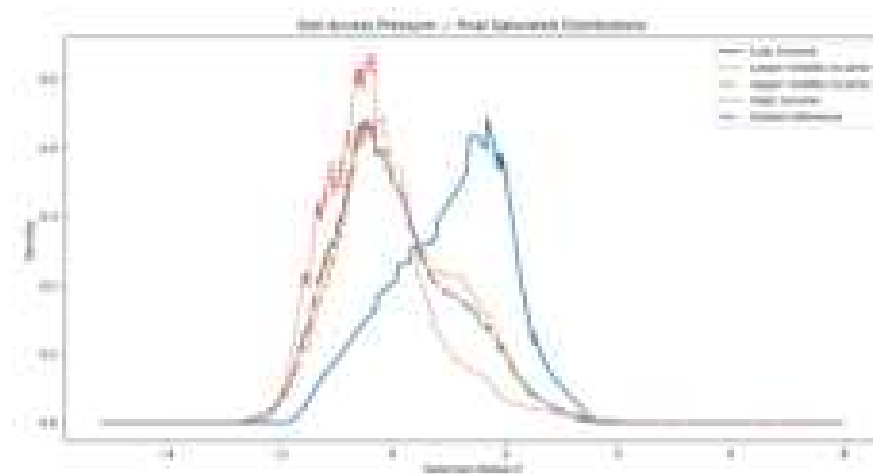


Figure B9. Construction Cost Pressure

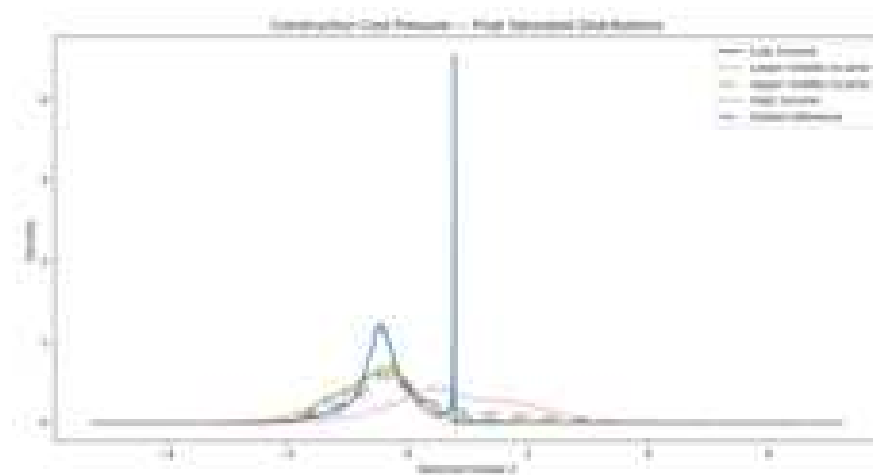


Figure B10. International Shipping Efficiency

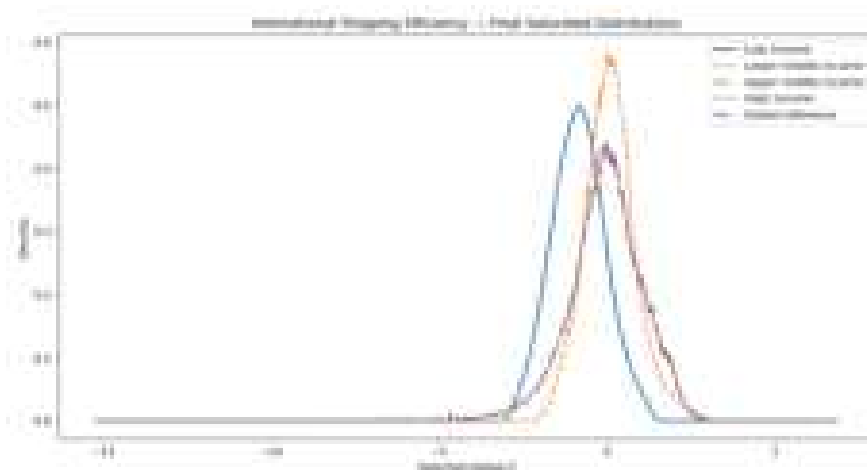


Figure B11. Electricity Cost Pressure

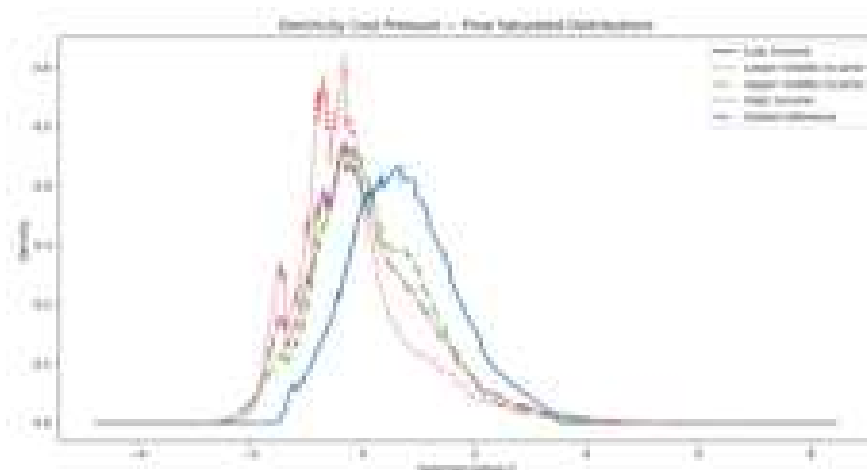


Figure B12. Tax Burden

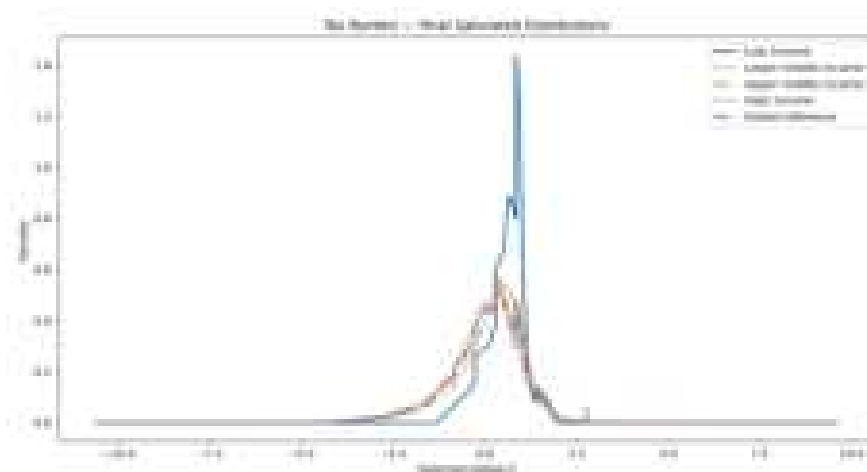


Figure B13. Political Stability

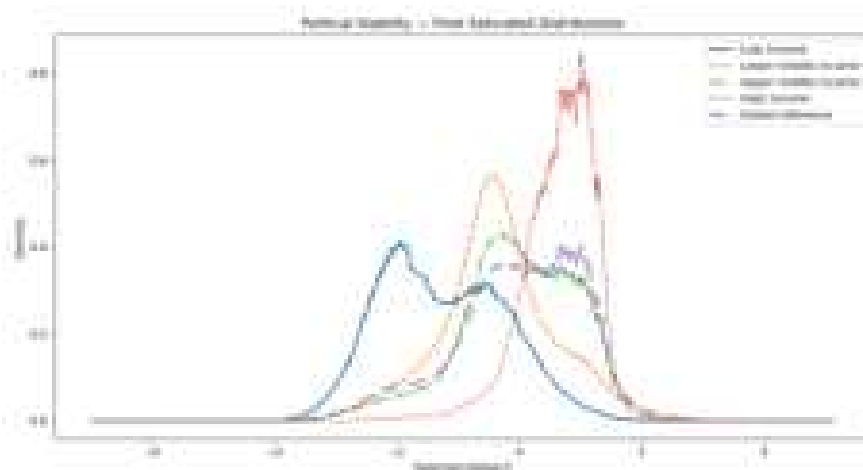


Figure B14. Regulatory Quality

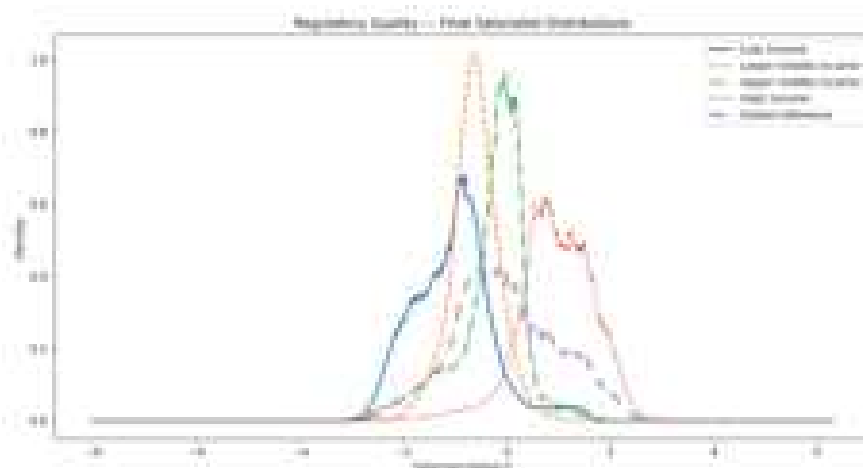


Figure B15. Macroeconomic Fragility

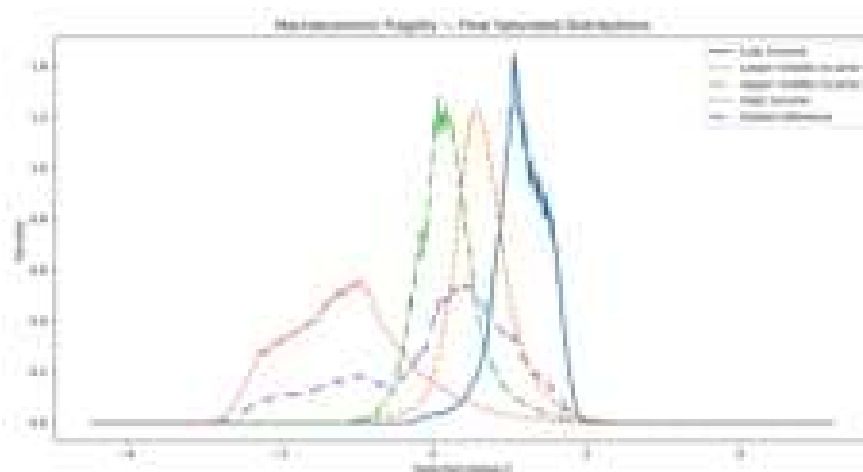


Figure B16. Inflation Risk

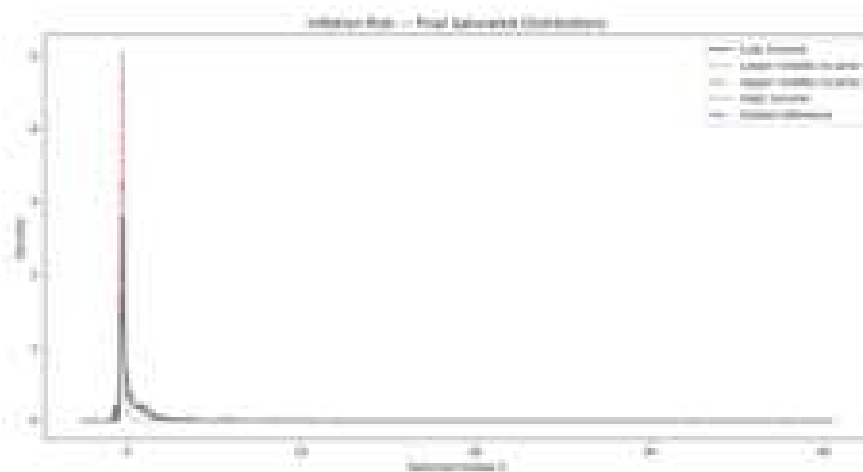


Figure B17. Institutional Quality

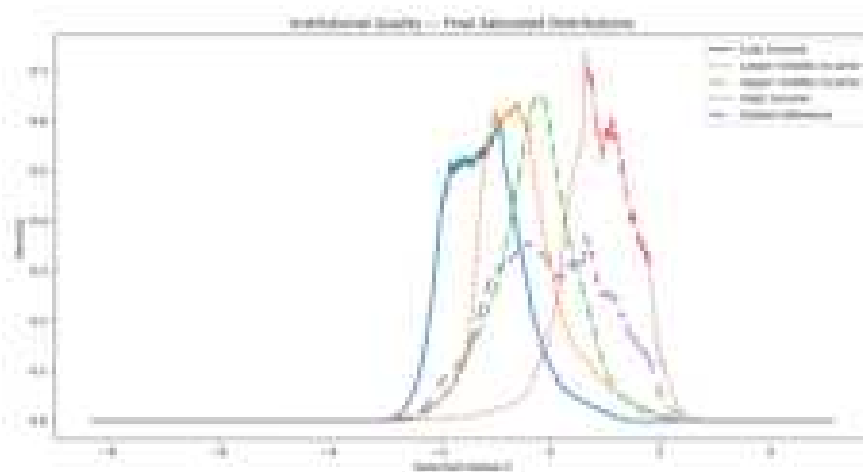


Figure B18. Corruption Risk

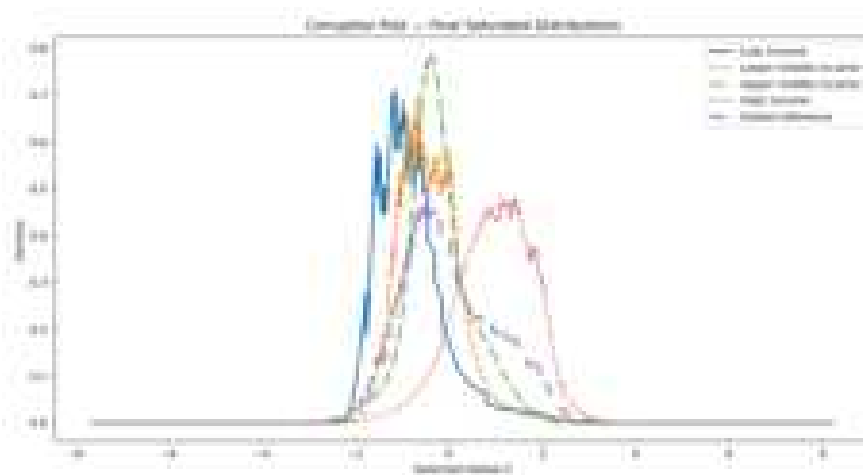


Figure B19. Logistics Disruption Risk

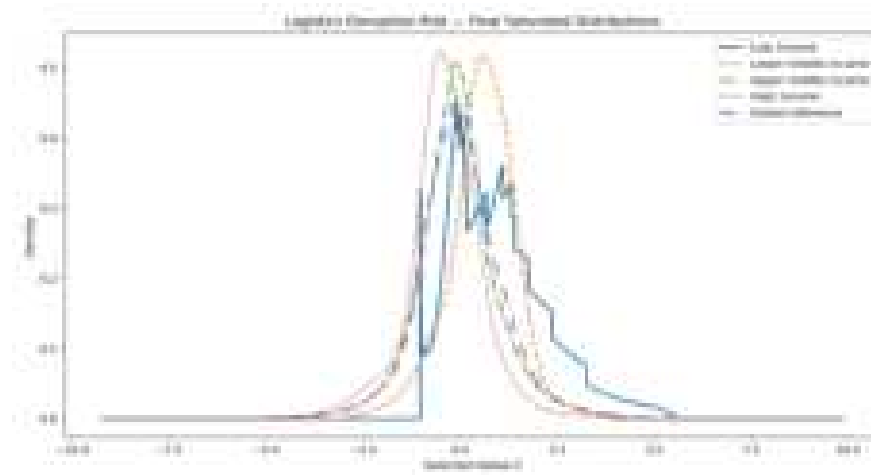


Figure B20. Construction Delay Risk

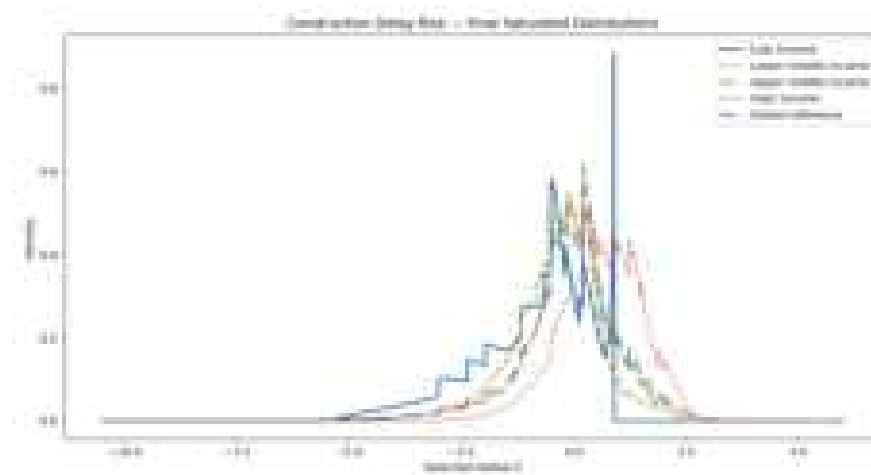


Figure B21. Financing Risk

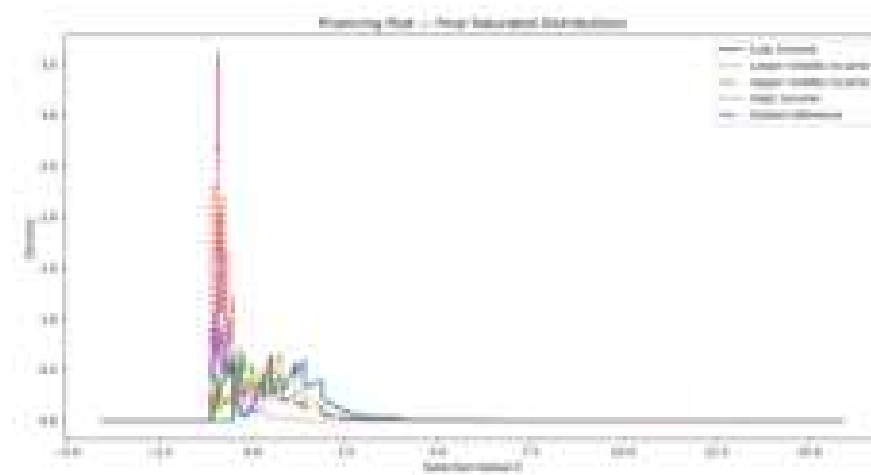
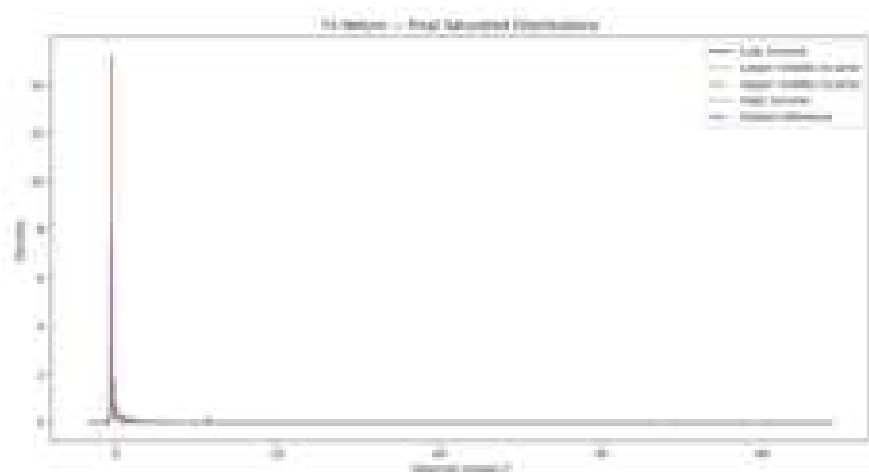
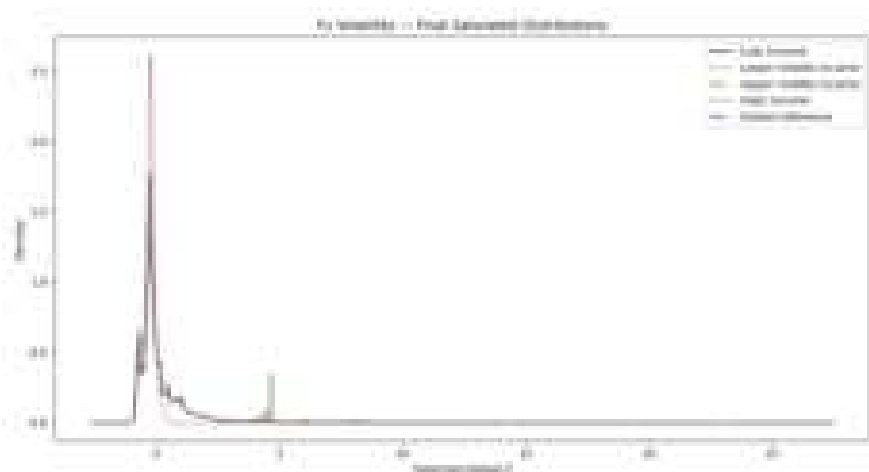


Figure B22. Expected FX Depreciation**Figure B23. FX Volatility**

The complete set demonstrates both meaningful income-group differentiation and substantial within-group heterogeneity. Household Consumption Capacity shows a clear development gradient, Inflation Risk displays strong tail asymmetry, and Metropolitan Customer Concentration shows extensive cross-group overlap. These patterns justify conditioning distributions by income group rather than treating the classifications as deterministic country scores.

Exhibit BAP04 - Final 23x23 Stochastic Dependence Matrix

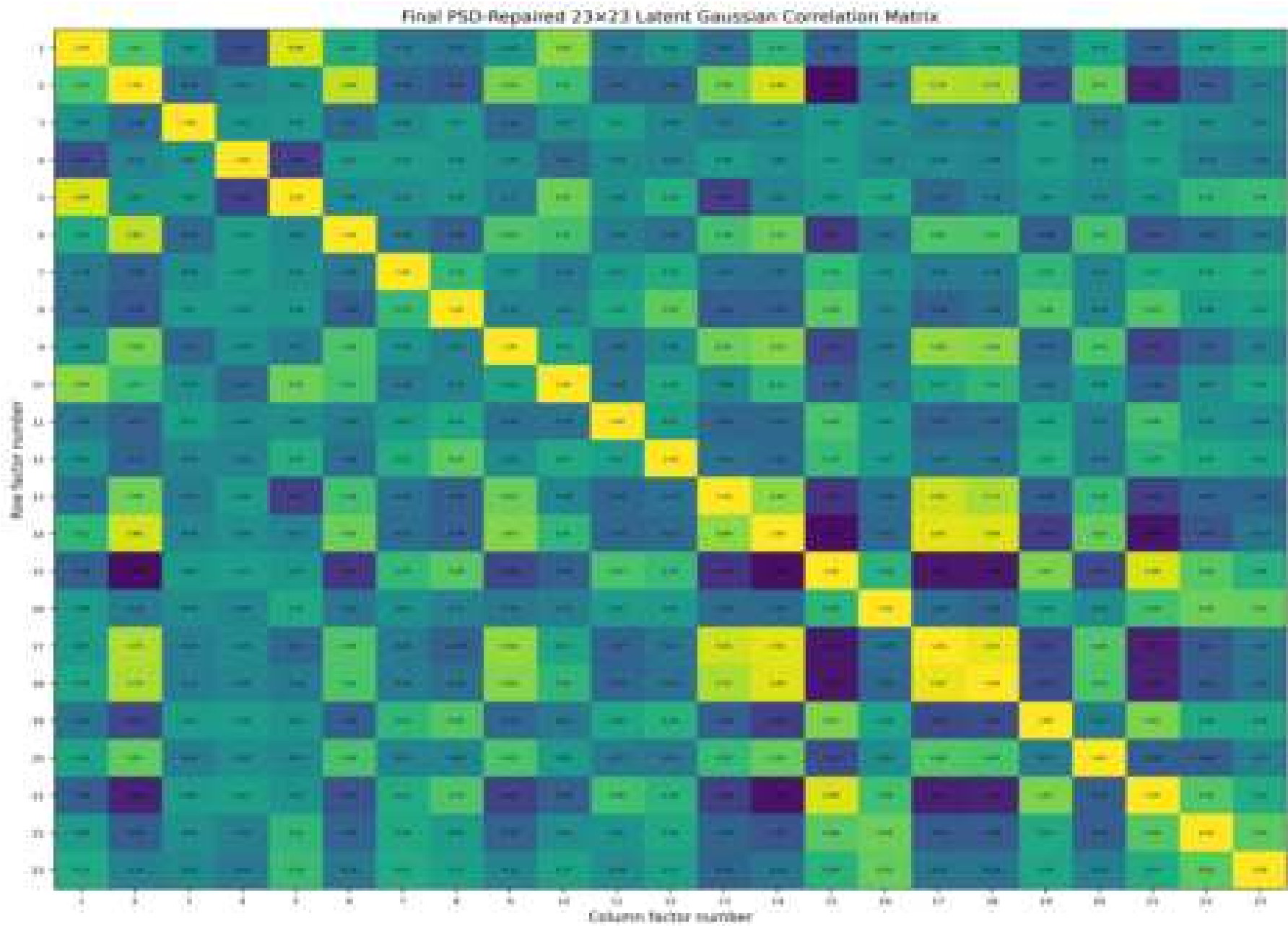


Exhibit BAP05 - Dependence Calibration and Audit Diagnostics

Diagnostic	Final result
Stochastic variables	23
Unique factor pairs	253
Strong evidence pairs	137 (54.2%)
Moderate evidence pairs	66 (26.1%)
Limited evidence pairs	50 (19.8%)
Mean Shared N	133.87
Median Shared N	140
Minimum Shared N	43
Maximum Shared N	210
Mean pairwise coverage	92.81%
Median pairwise coverage	96.67%
Median absolute source-year gap	1 year
Median pairwise P90 year gap	1 year
Bootstrap resamples per pair	1,000
Pairs with 95% interval excluding zero	158
Pairs retaining sign in $\geq 95\%$ resamples	170
Minimum eigenvalue before repair	-0.127385
Higham iterations	29
Minimum eigenvalue after repair	$\sim -3.67\text{e-}16$
Mean absolute off-diagonal adjustment	0.005241
Maximum absolute off-diagonal adjustment	0.041015
Joint validation draws / group	50,000

Pairwise Spearman correlations are estimated only from countries shared by the two variables. The evidence-quality flags summarize shared sample size, coverage, and source-year synchronization but do not automatically shrink or discard an observed coefficient. Bootstrap resampling measures stability. Because the assembled latent matrix was not PSD, Higham's (2002) nearest-correlation algorithm reconciles it into a valid matrix with relatively small average adjustments.

End-to-end validation after inverse-CDF transformation yields mean absolute error versus the PSD-implied Spearman target of approximately 0.0026-0.0036 across the four income groups. This confirms that the combination of the common latent dependence matrix and group-specific nonnormal marginals preserves the intended rank structure closely.

Appendix C - Monte Carlo Validation and Diagnostics

Appendix C retains compact controlled-direct and joint-correlated sensitivity summaries together with mechanism-level diagnostics for Stress, company residuals, inflation/FX, construction delay, and network support. All 23 factors were tested in the underlying sensitivity runs; full row-level outputs remain in the reproducibility archive rather than being repeated as dozens of tornado panels.

Exhibit C01 - Controlled Direct Sensitivity

Controlled direct sensitivity changes one factor while holding the broader path paired. It estimates the model's defined financial-transmission channel rather than the factor's total correlated importance.

Income group	Strategy	Three largest controlled-direct effects, signed Δ NPV (\$M)	Income group	Strategy	Three largest controlled-direct effects, signed Δ NPV (\$M)
Low income	Aggressive	Household Consumption Capacity (+60.4) Construction Cost Pressure (-49.4) Market Size (+40.2)	Upper middle income	Aggressive	Household Consumption Capacity (+300.6) Financing Risk (-223.5) FX Return / Depreciation (-153.0)
Low income	Moderate	Household Consumption Capacity (+43.1) Construction Cost Pressure (-35.7) Inflation Risk (-34.2)	Upper middle income	Moderate	Household Consumption Capacity (+246.1) Financing Risk (-175.6) FX Return / Depreciation (-109.3)
Low income	Organic	Household Consumption Capacity (+39.8) Inflation Risk (-36.8) Construction Cost Pressure (-32.0)	Upper middle income	Organic	Household Consumption Capacity (+226.8) Financing Risk (-165.7) FX Return / Depreciation (-99.1)
Lower middle income	Aggressive	Household Consumption Capacity (+135.2) Financing Risk (-94.0) FX Return / Depreciation (-63.4)	High income	Aggressive	Household Consumption Capacity (+753.0) Financing Risk (-465.4) FX Return / Depreciation (-447.2)
Lower middle income	Moderate	Household Consumption Capacity (+106.1) Financing Risk (-61.4) FX Return / Depreciation (-39.5)	High income	Moderate	Household Consumption Capacity (+660.2) Financing Risk (-418.2) FX Return / Depreciation (-386.0)
Lower middle income	Organic	Household Consumption Capacity (+95.7) Financing Risk (-46.4) Market Size (+31.5)	High income	Organic	Household Consumption Capacity (+637.3) Financing Risk (-401.9) FX Return / Depreciation (-365.1)

Note. Values are the three largest absolute controlled effects for each configuration; signs preserve the executed high-minus-low intervention. All 23 factors were tested.

Complete row-level output is retained in the V5 reproducibility archive.

Exhibit C02 - Joint Correlated Sensitivity

Joint sensitivity compares high versus low discounted factor-exposure paths while the full country system remains correlated. Resulting NPV differences are associations, not single-factor causal coefficients.

Income group	Strategy	Three largest joint P10-P90 associations, signed Δ NPV (\$M)	Income group	Strategy	Three largest joint P10-P90 associations, signed Δ NPV (\$M)
Low income	Aggressive	Financing Risk (-145.8) Macroeconomic Fragility (-128.1) Regulatory Quality (+122.6)	Upper middle income	Aggressive	Financing Risk (-675.3) Macroeconomic Fragility (-644.5) Regulatory Quality (+642.5)
Low income	Moderate	Financing Risk (-80.9) Inflation Risk (-79.0) Macroeconomic Fragility (-67.3)	Upper middle income	Moderate	Financing Risk (-552.1) Regulatory Quality (+522.6) Macroeconomic Fragility (-518.5)
Low income	Organic	Inflation Risk (-75.0) Financing Risk (-58.7) Macroeconomic Fragility (-45.5)	Upper middle income	Organic	Financing Risk (-519.7) Regulatory Quality (+487.9) Macroeconomic Fragility (-486.1)
Lower middle income	Aggressive	Financing Risk (-467.3) Macroeconomic Fragility (-453.3) Regulatory Quality (+427.7)	High income	Aggressive	Market Size (+839.7) Financing Risk (-742.9) Urban Market Availability (+630.5)
Lower middle income	Moderate	Financing Risk (-362.1) Macroeconomic Fragility (-346.2) Regulatory Quality (+327.4)	High income	Moderate	Market Size (+735.6) Financing Risk (-662.7) Urban Market Availability (+547.8)
Lower middle income	Organic	Financing Risk (-333.5) Macroeconomic Fragility (-315.0) Regulatory Quality (+298.0)	High income	Organic	Market Size (+709.7) Financing Risk (-643.7) Urban Market Availability (+527.2)

Note. Joint values are associations within the complete correlated country state, not single-factor causal coefficients. All 23 factors were evaluated; complete row-level output

is retained in the V5 reproducibility archive.

Exhibit C03 — Stress, Residual, Delay, and Network Diagnostics

Figure C1. Stress Ablation — P10 NPV

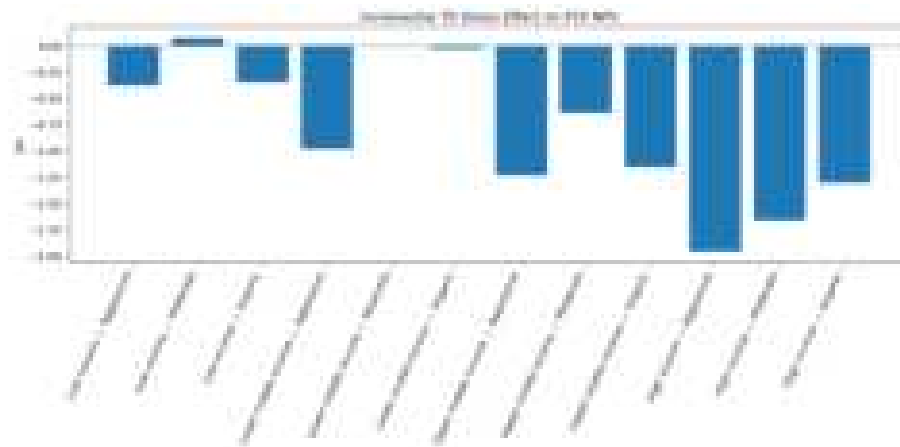


Figure C2. Company Residual Dependence Robustness

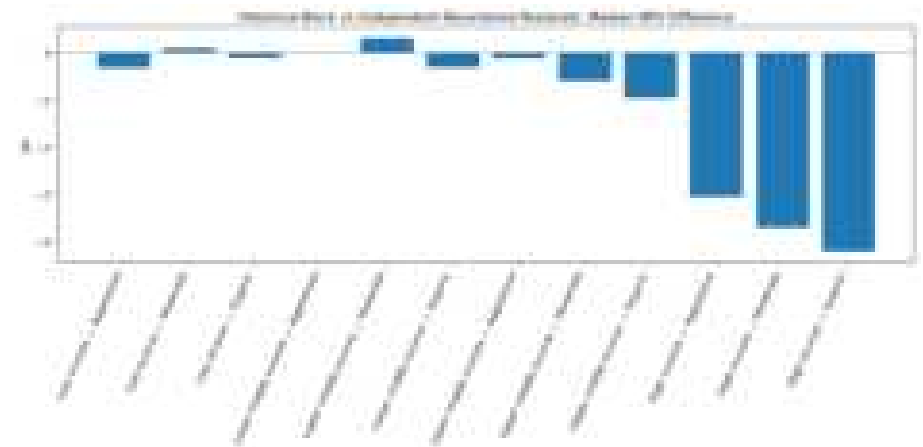


Figure C3. Executed Construction-Delay Mapping

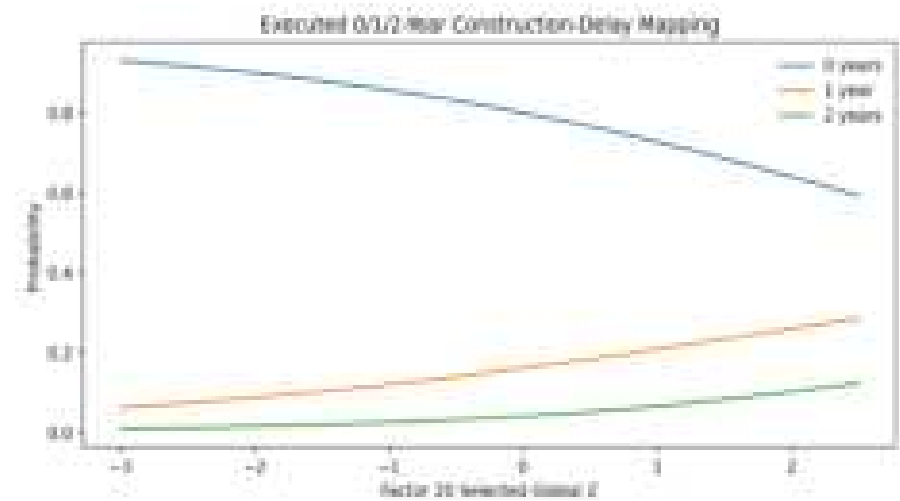
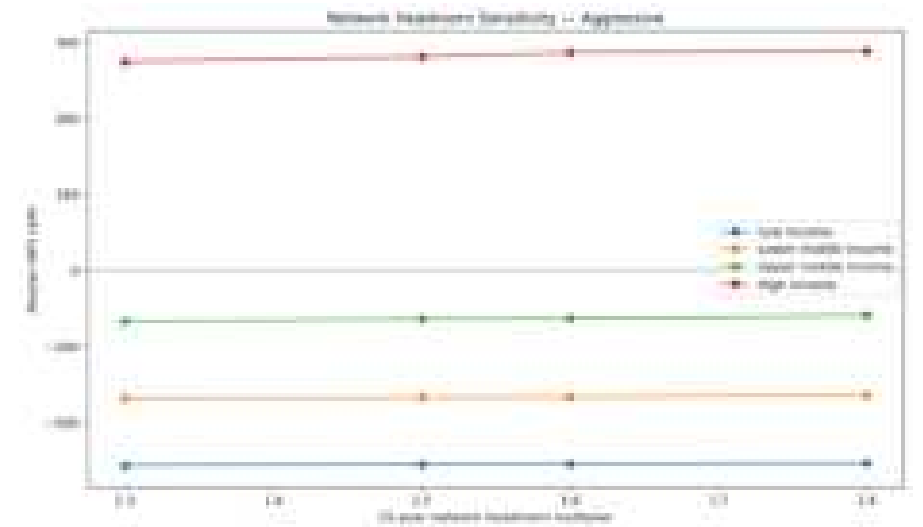


Figure C4. Network Headroom Sensitivity

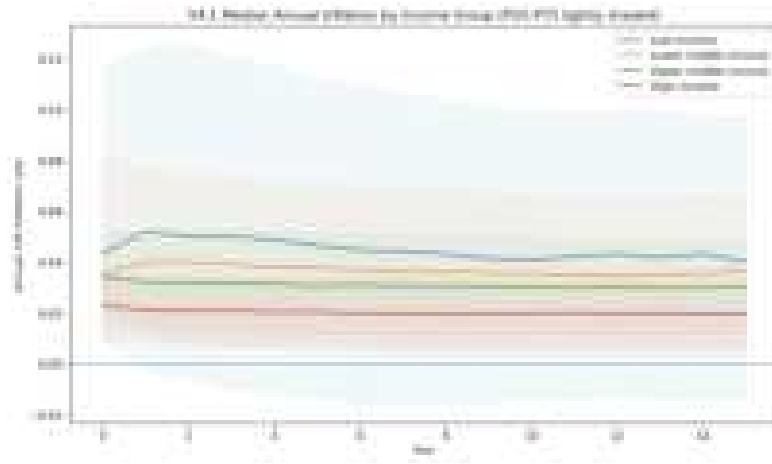


Note. Stress is a synchronized mechanism ablation; the residual diagnostic compares independent versus historical-year block sampling; construction delay uses frozen V5's direct 0/1/2-year mapping; network sensitivity tests 1.30x-1.80x headroom.

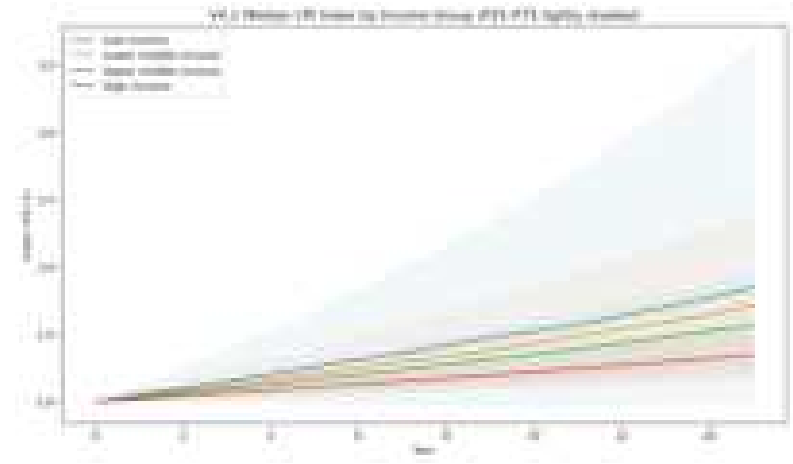
The Stress mechanism remains modest relative to the broad NPV distributions and changes risk-state innovations rather than applying a direct crisis haircut. Historical-year block residual sampling changes median NPV by only a few million dollars in the mechanism audit. Construction delay shifts real cash-flow timing and can have a positive NPV sign in weak projects because postponing value-destructive investment can improve value. Network headroom changes valuation magnitude but does not overturn the broad decision zones for the six-store experiment.

Exhibit C04 — Inflation, CPI, FX, and Relative-Price Diagnostics

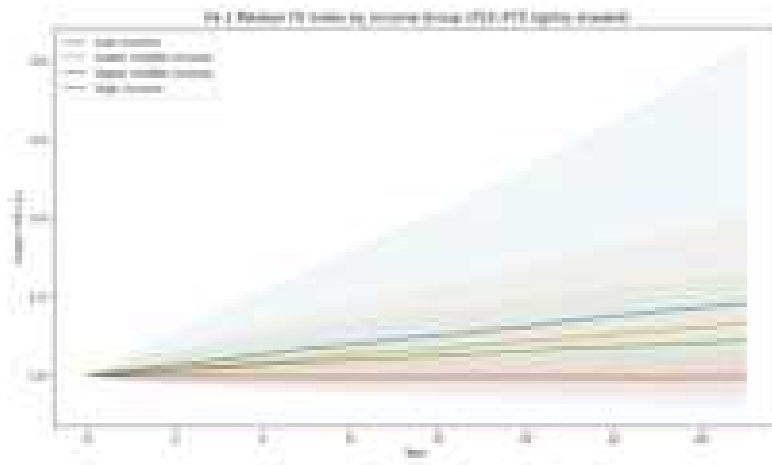
Median annual inflation



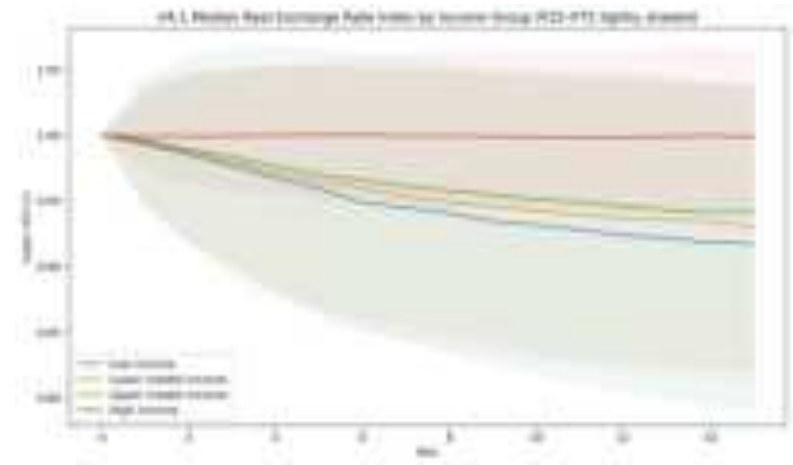
Median CPI index



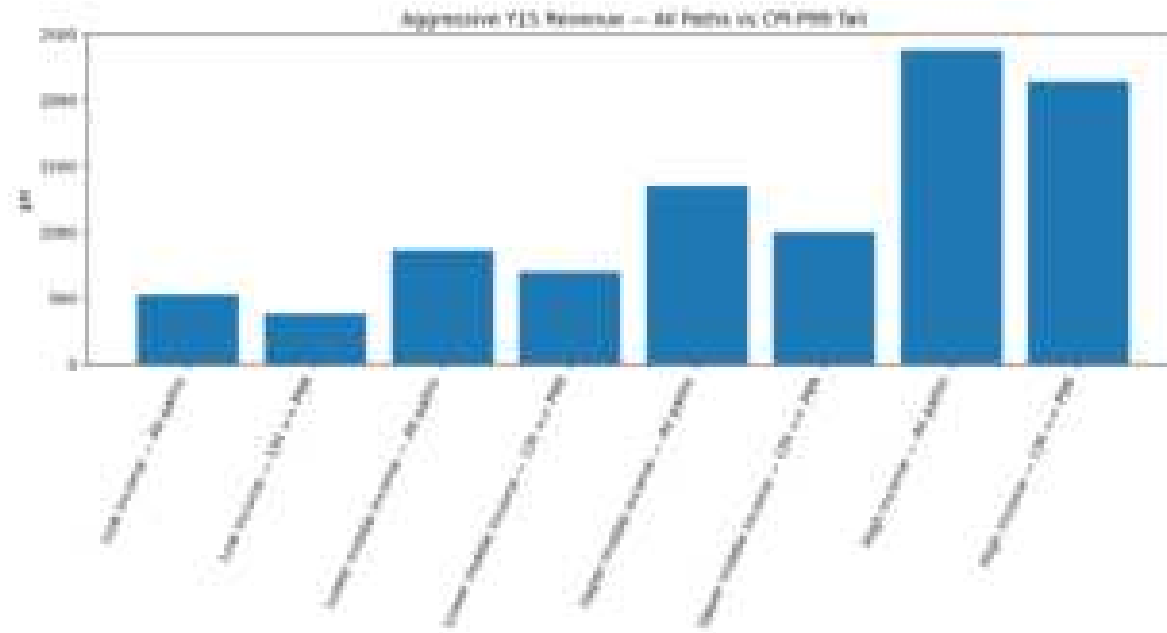
Median FX index



Median real exchange rate



Note. Year-15 central inflation converges toward the forward anchors while empirical dispersion and FX innovation remain active. The real-exchange-rate panel is a long-horizon consistency diagnostic, not a hard target of one.

Figure C5.*Aggressive Year-15 Revenue Under Extreme CPI / FX Tail States*

Note. The tail diagnostic shows why strong local nominal revenue does not necessarily translate into strong sponsor-currency value.

Appendix D - Detailed Scenario Results

Appendix D reports the frozen V5 10,000-path valuation, return, execution, financing, operating, terminal, and annual trajectory evidence underlying Section 7.

Exhibit D01 - Complete Valuation and Return Results

Income group	Strategy	P10 NPV	Median NPV	P90 NPV	P(NPV>0)	Median IRR	Eff. WACC	Median MOIC	Revenue CAGR	P(Payback)
Low	Aggressive	-323.2	-268.9	-170.3	1.4%	-3.0%	20.6%	0.65x	10.3%	16.7%
Low	Moderate	-161.8	-139.8	-107.1	1.6%	-6.0%	20.6%	0.56x	12.1%	7.5%
Low	Organic	-53.2	-42.2	-27.3	1.7%	-0.6%	20.6%	0.87x	9.2%	19.5%
Lower-middle	Aggressive	-286.0	-166.7	107.2	17.7%	6.5%	17.6%	1.91x	10.0%	57.5%
Lower-middle	Moderate	-144.6	-107.2	90.9	17.9%	5.3%	17.6%	1.51x	13.6%	42.1%
Lower-middle	Organic	-46.0	-28.1	23.0	16.3%	5.8%	17.6%	1.51x	11.4%	55.4%
Upper-middle	Aggressive	-281.8	-71.3	353.7	38.4%	12.2%	15.3%	3.28x	10.1%	72.7%
Upper-middle	Moderate	-143.1	-50.0	302.9	39.2%	12.4%	15.3%	3.10x	14.1%	65.6%
Upper-middle	Organic	-43.8	-14.2	120.7	37.5%	11.7%	15.3%	2.07x	14.0%	65.4%
High	Aggressive	-317.7	257.0	833.2	69.8%	17.6%	10.4%	5.77x	11.3%	76.8%
High	Moderate	-159.5	226.9	760.2	70.7%	18.6%	10.4%	5.92x	15.7%	73.3%
High	Organic	-46.0	69.6	418.4	74.2%	19.1%	10.4%	3.59x	16.6%	76.8%

Note. Monetary values are millions of U.S. dollars. Effective WACC is a path-equivalent reporting hurdle.

Exhibit D02 - Execution and Financing Results

Income group	Strategy	P(6/6)	Median completed	P(financing delay)	Median delay years	Median sponsor capital	Median peak debt	Strict full success
Low	Aggressive	100.0%	6	0.0%	0	318.3	0.0	1.3%
Low	Moderate	25.1%	4	100.0%	30	159.4	7.1	1.5%
Low	Organic	0.1%	1	100.0%	71	53.4	0.0	0.1%
Lower-middle	Aggressive	100.0%	6	0.0%	0	286.8	0.0	17.0%
Lower-middle	Moderate	67.8%	6	96.9%	12	145.6	24.3	17.7%
Lower-middle	Organic	3.1%	2	100.0%	66	49.4	0.0	3.1%
Upper-middle	Aggressive	100.0%	6	0.0%	0	286.8	0.0	37.1%
Upper-middle	Moderate	79.6%	6	89.1%	5	146.6	31.9	38.7%
Upper-middle	Organic	9.2%	2	100.0%	58	50.1	0.0	9.1%
High	Aggressive	100.0%	6	0.0%	0	322.8	0.0	69.4%
High	Moderate	81.9%	6	74.4%	2	165.5	55.8	70.6%
High	Organic	19.9%	3	100.0%	51	57.0	0.0	19.9%

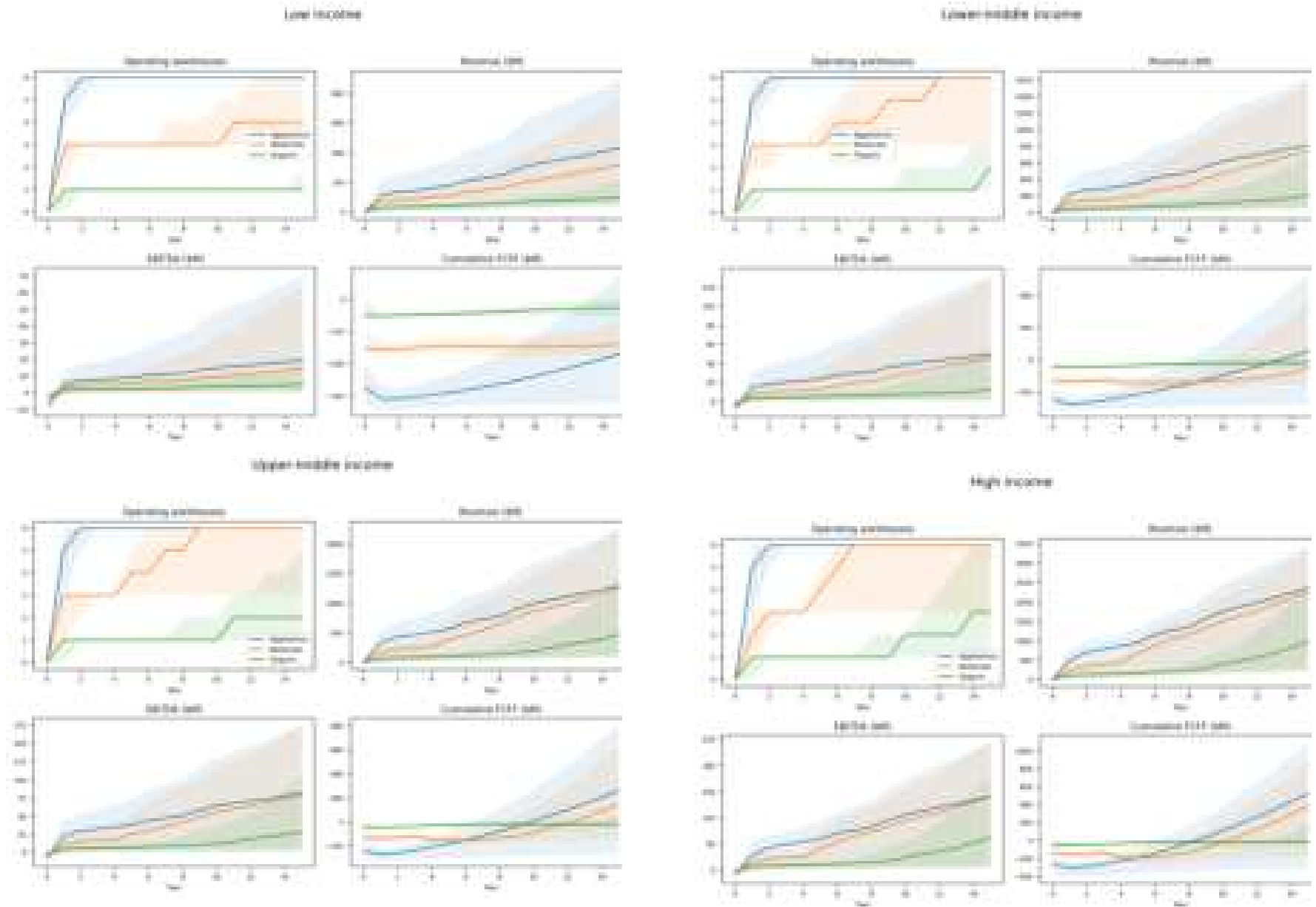
Note. Sponsor / External Capital is the report term for the executable's historical Parent Funding output.

Exhibit D03 - Operating and Terminal Economics

Income group	Strategy	Y15 revenue	Y15 Gross Profit (\$M)	Y15 EBITDA	EBITDA margin	Y15 FCFF	Cumulative FCFF	Terminal EV	PV terminal EV
Low	Aggressive	430.8	38.5	19.6	5.20%	12.9	-172.7	58.1	3.3
Low	Moderate	319.8	28.7	14.5	5.11%	7.1	-140.1	43.4	2.5
Low	Organic	99.4	9.2	4.9	5.15%	2.5	-26.0	15.0	0.9
Lower-middle	Aggressive	810.1	80.0	49.5	6.45%	30.4	52.9	194.5	15.7
Lower-middle	Moderate	743.3	74.3	45.5	6.42%	28.8	-51.8	179.2	14.5
Lower-middle	Organic	204.3	20.2	12.1	6.30%	4.2	-20.3	47.9	4.1
Upper-middle	Aggressive	1295.1	134.5	81.2	6.57%	51.3	257.1	379.9	40.5
Upper-middle	Moderate	1253.1	130.1	78.7	6.56%	50.2	139.7	367.3	39.1
Upper-middle	Organic	466.1	48.8	28.7	6.44%	6.4	-19.4	137.9	15.1
High	Aggressive	2350.6	277.1	140.8	6.65%	89.0	523.2	1038.3	225.9
High	Moderate	2251.4	265.5	139.5	6.64%	88.3	379.8	1027.8	223.4
High	Organic	1000.7	117.5	62.3	6.55%	11.0	-14.8	458.4	101.1

Note. Monetary values are millions of U.S. dollars. Gross profit is defined as total project revenue less merchandise cost; membership revenue has no separate merchandise COGS line in the project pro forma. Terminal EV is a DCF continuation value, not an externally imposed comparable-company multiple.

Exhibit D04 - Annual Trajectory Composites



Note. Each composite shows operating warehouses, revenue, EBITDA, and cumulative FCFF using frozen V5 annual quantiles.

Appendix E - Data Sources and Provenance

Appendix E traces principal model inputs from provider and indicator through transformation and final V5 use. The registry distinguishes direct calibration from mechanism literature, structural priors, and development/non-executed provenance.

Exhibit E01 - Master Data Sources and Provenance Registry

Factor	Provider	Series / source ID	Raw evidence	Selected pre-Z transform	Final V5 use
F01 Market Size	World Bank WDI (World Bank, 2026c)	NY.GDP.MKTP.PP.CD	PPP GDP	Natural log (ln)	Network support
F02 Household Consumption Capacity	World Bank WDI (World Bank, 2026c)	NE.CON.PCAP.PP.CD	PPP consumption/capita	Natural log (ln)	Revenue, membership, support
F03 Expected Market Growth	World Bank WDI (World Bank, 2026c)	NY.GDP.MKTP.KD.ZG	GDP growth %	Raw	Real growth, support growth
F04 Metropolitan Customer Concentration	World Bank WDI (World Bank, 2026c)	EN.URB.LCTY.UR.ZS	Largest-city share	Raw	Demand accessibility
F05 Urban Market Availability	World Bank WDI (World Bank, 2026c)	SP.URB.TOTL	Urban population	Natural log (ln)	Demand accessibility
F06 Labor Cost Pressure	ILO ILOSTAT (International Labour Organization, n.d.-b)	EAR_EHRA_SEX_EC_O_CUR_NB_A	PPP hourly earnings	Natural log (ln)	Labor SG&A / staffing
F07 Labor Underutilization	ILO ILOSTAT (International Labour Organization, n.d.-a)	LU4	Underutilization %	Raw	Staffing friction
F08 Site Access Pressure	World Bank B-READY (World Bank, 2026a)	Land-access measure	Index	Raw	Land/site cost
F09 Construction Cost Pressure	World Bank ICP (World Bank, 2025a)	1501200:CONSTRUCTION	Construction price level	Natural log (ln)	Hard-building CapEx
F10 International Shipping Efficiency	World Bank LPI 2.0 (Arvis et al., 2026)	Maritime partners	Count	Natural log (ln)	MCR, inventory, equipment
F11 Electricity Cost Pressure	World Bank B-READY (World Bank, 2026a)	Commercial electricity bill	Cost	Raw	Nonlabor SG&A
F12 Tax Burden	PwC via Damodaran (Damodaran, 2026a)	Country tax-rate dataset	Corporate tax %	Raw	Cash tax
F13 Political Stability	World Bank WGI (World Bank, 2025b)	PV.EST	Governance estimate	Raw	Dependence / Stress
F14 Regulatory Quality	World Bank WGI (World Bank, 2025b)	RQ.EST	Governance estimate	Raw	SG&A / development friction
F15 Macroeconomic Fragility	Fund for Peace FSI (Fund for Peace, n.d.)	E1 Economic Decline	Index	Raw	Dependence / Stress
F16 Inflation Risk	World Bank WDI (World Bank, 2026c)	FP.CPI.TOTL.ZG	CPI inflation %	Raw	Forward CPI system
F17 Institutional Quality	World Bank WGI (World Bank, 2025b)	RL.EST	Governance estimate	Raw	MCR, SG&A, inventory

Factor	Provider	Series / source ID	Raw evidence	Selected pre-Z transform	Final V5 use
F18 Corruption Risk	Transparency International (2026)	CPI 2025	Score	Raw	Operating / development friction
F19 Logistics Disruption Risk	World Bank LPI 2.0 (Arvis et al., 2026)	CT_DT_M	Dwell-time dispersion	Natural log (ln)	MCR, inventory, Stress
F20 Construction Delay Risk	World Bank B-READY (World Bank, 2026a)	Permit time	Time	Natural log (ln)	Executed 0/1/2-year mapping
F21 Financing Risk	Damodaran (2026b)	Country default spread	% spread	Raw	WACC / Project Debt / AP / Stress
F22 Expected FX Depreciation	World Bank WDI (World Bank, 2026c)	PA.NUS.FCRF	LCU/USD	Raw	Central FX state
F23 FX Volatility	BIS XRU (Bank for International Settlements, n.d.)	Bilateral exchange rates	Monthly FX	Raw	FX innovation scale

Note. “Selected pre-Z transform” reports the transformation actually selected before global standardization and empirical marginal calibration. “Natural log (ln)” means the positive raw variable was log-transformed before its Global Z score was constructed; “Raw” means no additional natural-log transformation was selected. Factor 23 is built from monthly FX log returns, but the resulting annualized volatility measure enters the country-distribution calibration on its raw constructed scale. F20 uses B-READY as the executed construction-delay state; World Bank FMIS is retained only as development/non-executed provenance and does not calibrate the frozen V5 delay-duration mechanism.

Exhibit E01 (continued) - Company, Capital Markets, Priors, and Non-Executed Provenance

Input / topic	Source context	Final specification / transformation	Evidence role
Costco company residuals	Costco 10-Ks, FY2016–FY2025 (Costco Wholesale Corporation, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025)	Six bounded/recentered residual families	Direct empirical calibration
Warehouse-age productivity	Costco FY2025 cohort/calendar evidence (Costco Wholesale Corporation, 2025)	Audited 10-year deterministic curve	Direct empirical calibration
Risk-free term structure	Federal Reserve GSW zero-coupon curve (Board of Governors of the Federal Reserve System, n.d.; Gürkaynak et al., 2006)	Annual forward risk-free rates	Direct market calibration
Mature-market ERP / country risk	Damodaran January 2026 (Damodaran, 2026b, 2026c)	ERP 4.23%; country DS/CRP relationships	Direct market calibration
Corporate debt spread	ICE Data Indices via FRED (ICE Data Indices, LLC, n.d.)	0.79% central OAS anchor	Direct market calibration
PPP correction	PPP mean-reversion mechanism (Frankel & Rose, 1995)	15% annual correction	Literature-informed structural prior
Network headroom / post-support	Retail-network mechanism literature (Einav et al., 2021; Guler, 2018; Holmes, 2008; Lim et al., 2021) + model design	1.50x headroom; 75% at U=2	Researcher structural prior
Moderate credit policy	Credit and financing literature (Nini et al., 2009; S&P Global Ratings, 2024; Sufi, 2009) + model design	3.0x / 6.0x / 15% tests	Researcher structural prior
Terminal growth	Researcher/model design	2.5% with 1 pp WACC-g floor	Researcher structural prior
FMIS positive delay duration	World Bank FMIS July 2026 (World Bank, 2026b)	NOT EXECUTED in frozen V5	Development / non-executed provenance

Note. The provenance chain is provider -> dataset/period -> raw measurement -> transformation or explicit prior -> defined financial role -> frozen V5 executable -> validation -> final scenario outputs. Development evidence is never presented as executed calibration when the frozen code does not use it.

Appendix F - AI Use Note

Generative artificial intelligence, primarily OpenAI's ChatGPT, was used throughout this independent study as a research, technical, and writing-support tool. Because the project involved the development of a relatively complex Monte Carlo valuation framework, AI was used iteratively rather than for a single isolated task. Its role included conversational brainstorming, organizing research questions and model architecture, assisting with Python code development and debugging, reviewing model logic and diagnostic outputs, helping document methodological decisions, identifying areas requiring additional source verification, and supporting drafting, editing, and formatting of the final report.

AI was also used during the model-development process to help evaluate whether implemented behavior was consistent with the intended financial and economic logic. This included discussion of the Country Screening Tool, AHP structure, empirical marginal distributions, stochastic dependence, temporal country-state processes, inflation and foreign exchange, warehouse-age productivity, network support, capital markets, financing continuation, valuation outputs, and validation procedures. When potential inconsistencies were identified, the underlying model files, calibration evidence, or source material were reviewed before deciding whether a numerical change, documentation correction, or no change was appropriate.

AI-generated statements were not treated as empirical evidence or cited as substitutes for original sources. Company information, country indicators, market data, methodological references, and academic literature appearing in the report were traced to and checked against the underlying sources before inclusion. Literature was used to support methods, mechanisms, and interpretation without assuming that outside studies had estimated Costco-specific financial coefficients where they had not. Numerical assumptions that could not be directly estimated from available evidence were identified as structural or literature-informed priors rather than presented as externally established facts.

The final model was not accepted solely because its code executed successfully. Model development included deterministic reconstruction, consistency checks, stochastic validation, sensitivity

analysis, economic-plausibility review, financing and execution invariants, reproduction testing, and repeated audits of the relationship between documentation and the frozen executable specification. AI assisted with portions of this review process, but final judgments regarding model architecture, calibration choices, revisions, interpretation of findings, and research conclusions were made by the author.

The author remained responsible for the research design, AHP judgments, source selection, model assumptions, interpretation of results, written conclusions, and final submitted work. Generative AI therefore functioned as an iterative research and technical support tool rather than as an independent source of evidence or substitute for authorial judgment.